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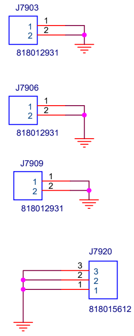
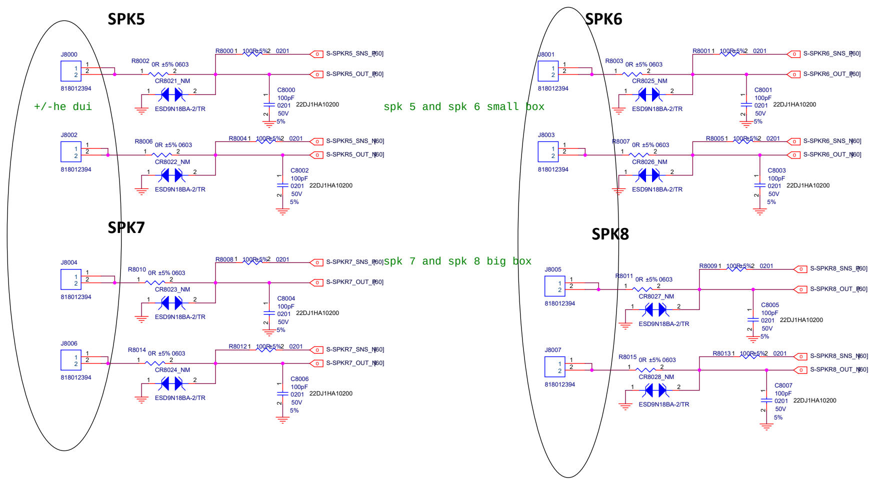
Size A3	Document Number CONTENT	Rev
Date:	Thursday, May 27, 2021	Sheet

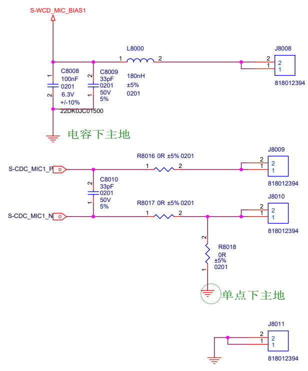


Title F1 MAIN		
Size A3	Document Number Reserved(for PA)	Rev
Date:	Thursday, May 27, 2021	Sheet

SPKEAKER



MAIN MIC



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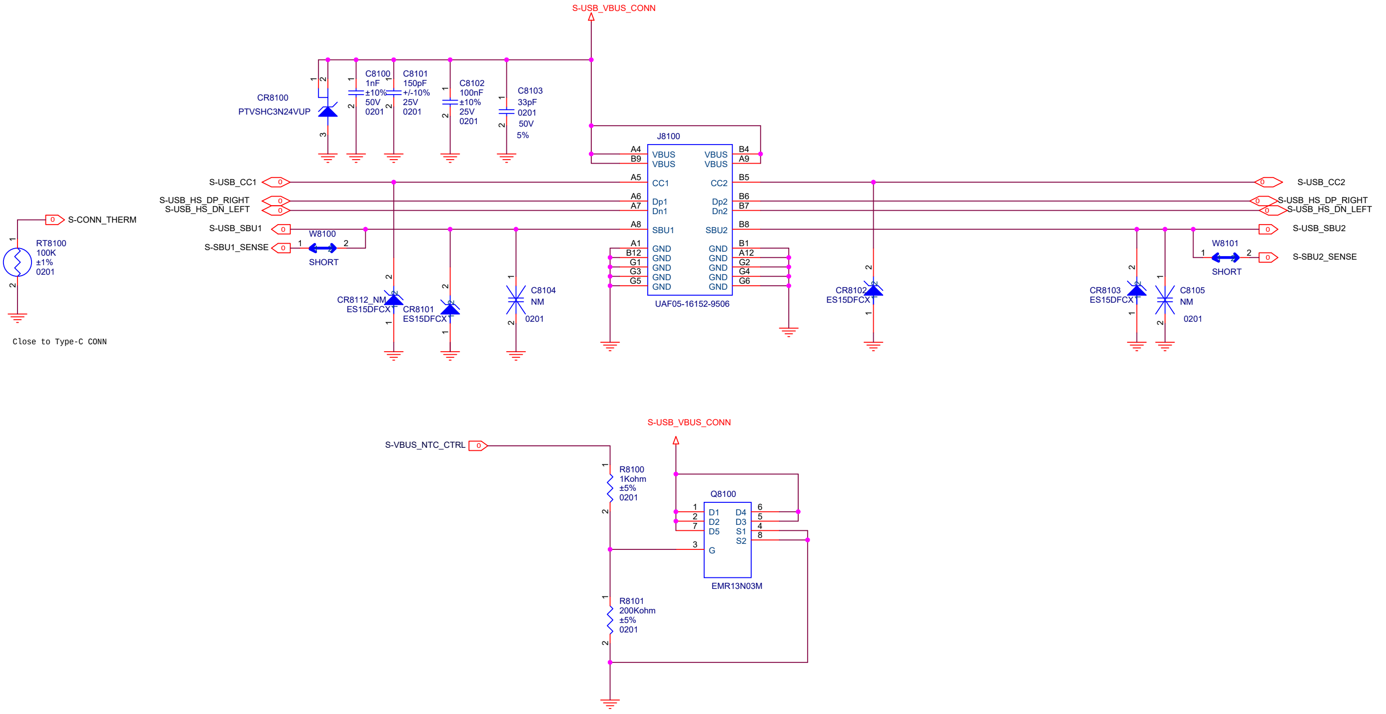
Size: A2 | Document Number: SMB1390&WL POWER | Rev

Date: Thursday, May 27, 2021 | Sheet



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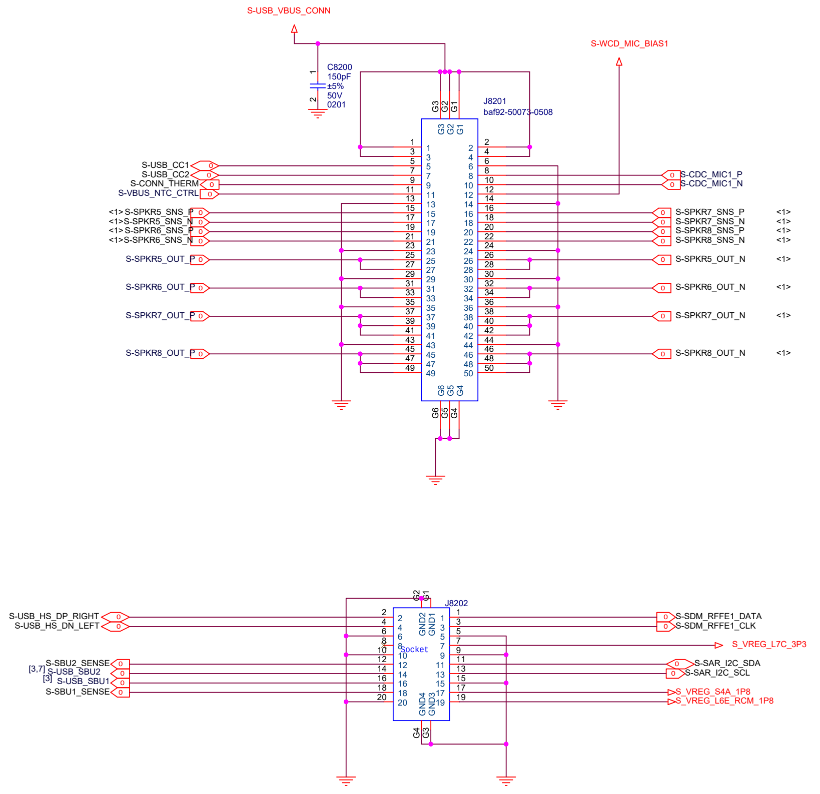
Size A3	Document Number <Doc>	Rev
Date: Thursday, May 27, 2021	Sheet 0	4



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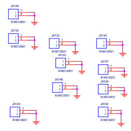
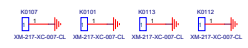
Size A3	Document Number <Doc>	Rev
Date: Thursday, May 27, 2021	Sheet 0 of 4	

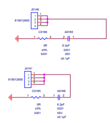
Title <Title>		
Size A1	Document Number <Doc>	Rev <RevCode>
Date: Thursday, May 27, 2021	Sheet 1 of 1	

GND

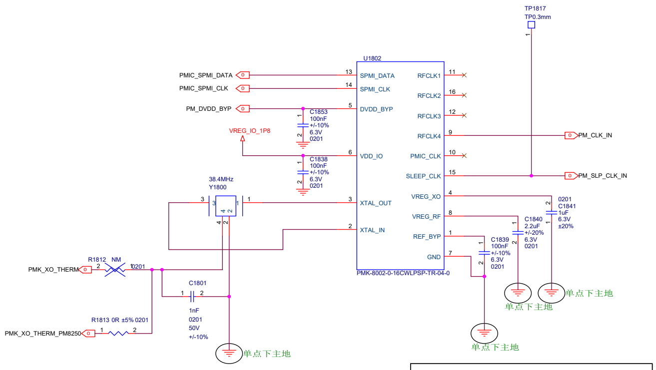




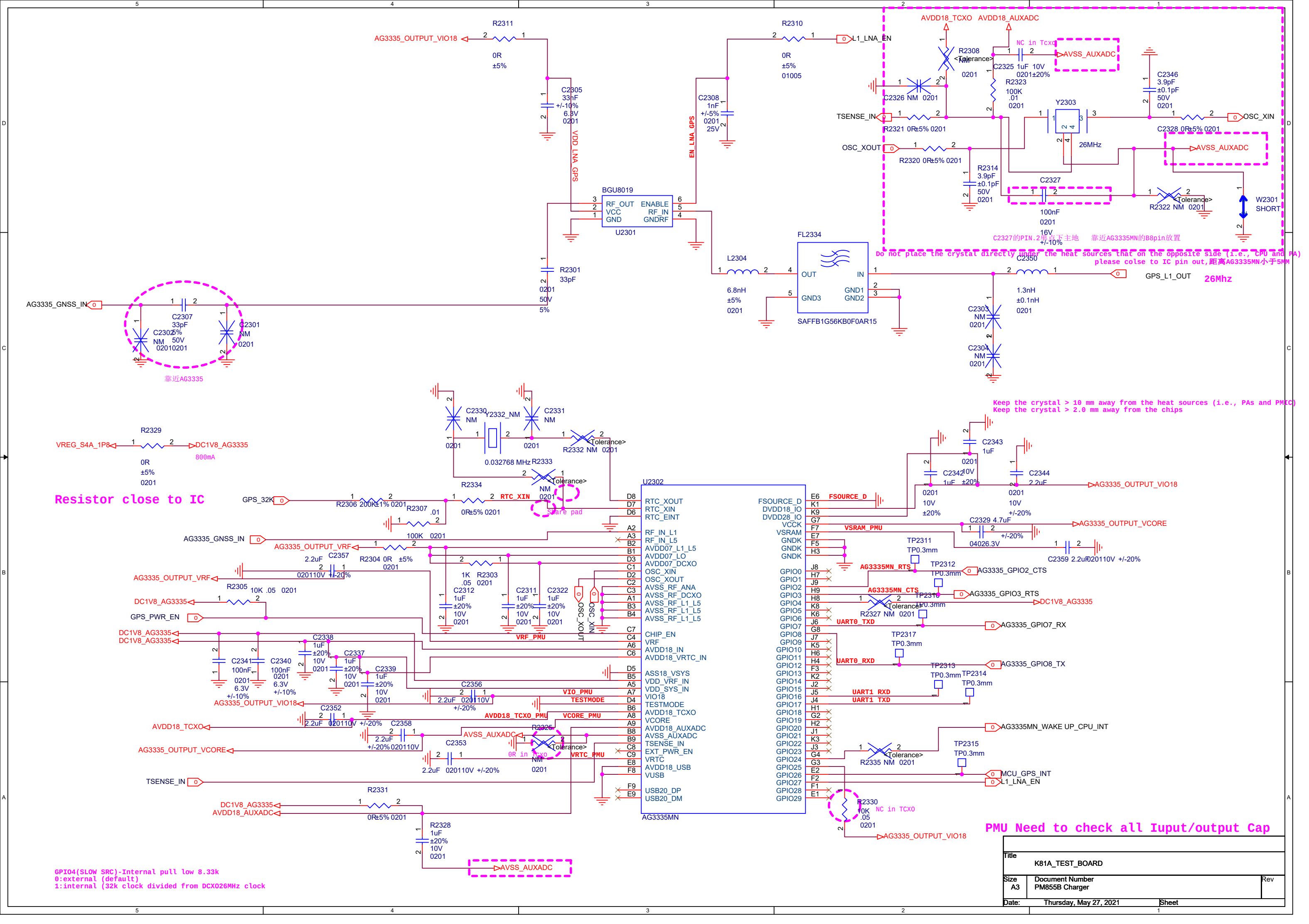
SAR Sensor ADUX1050BCBZ

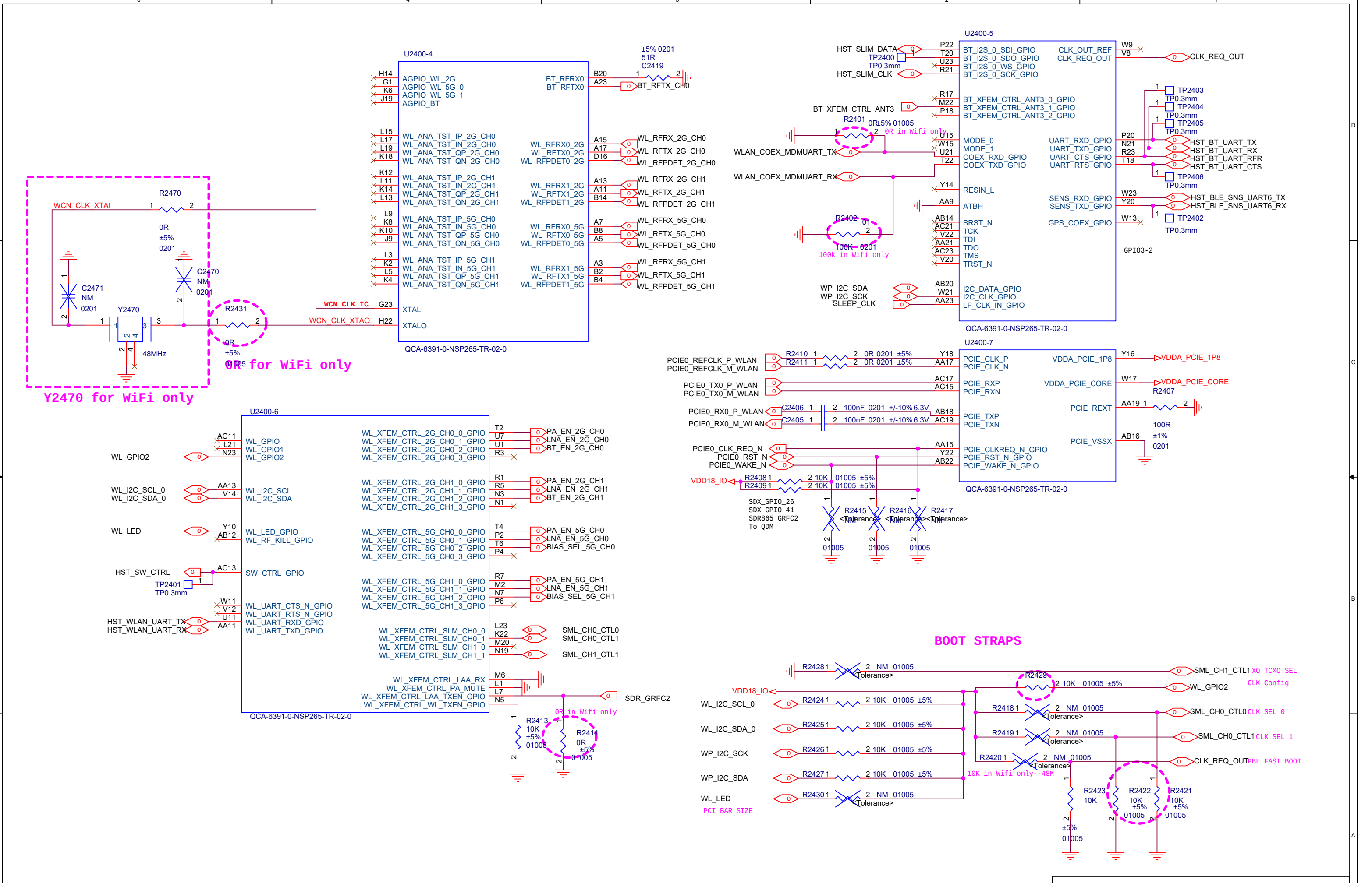


Title	K81A_TEST_BOARD	
Size A0	Document Number PM855B Charger	Rev
Date: Thursday, May 27, 2021	Sheet	

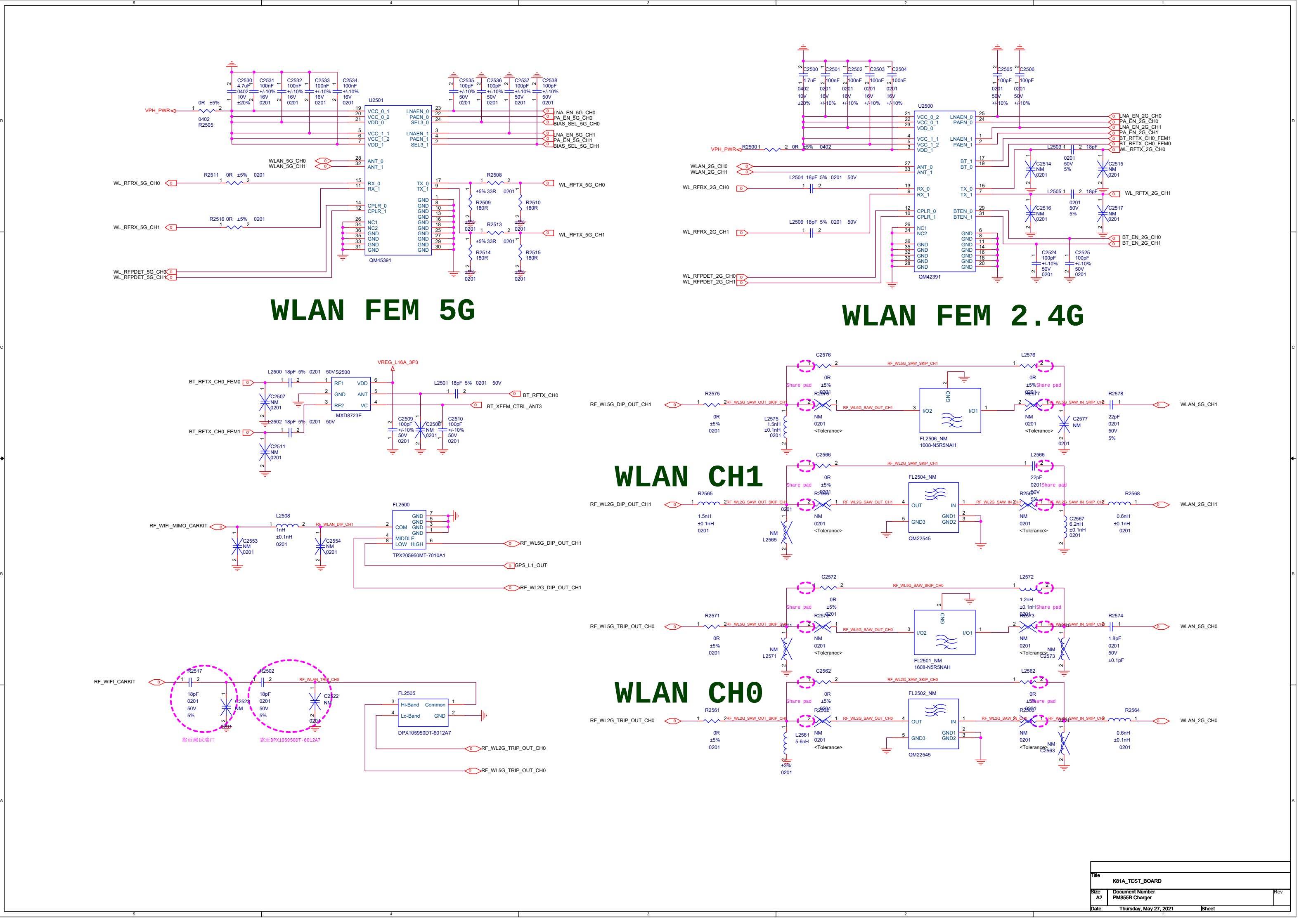


Title	K81A_TEST_BOARD	
Size A2	Document Number Reserved(for 5G-WTR I)	Rev
Date: Thursday, May 27, 2021	Sheet	

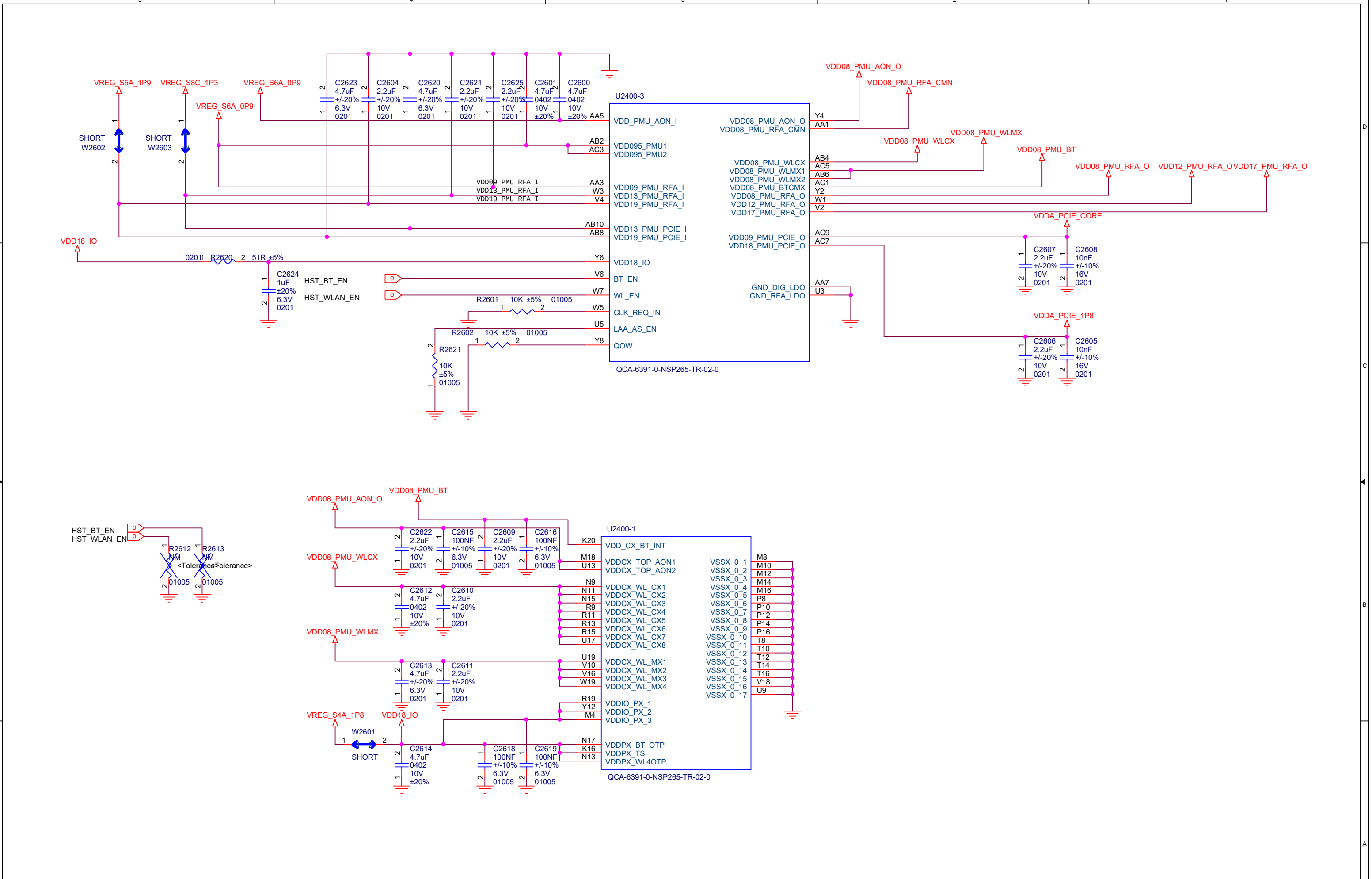

Resistor close to IC
PMU Need to check all Iuput/output Cap
Keep the crystal > 10 mm away from the heat sources (i.e., PAs and PMIC)
Keep the crystal > 2.0 mm away from the chips
Do not place the crystal directly under the heat sources that on the opposite side (i.e., CPU and PA)
please colse to IC pin out,距离AG3335MN小于5MM
26Mhz
C2327的PIN.2单点下主地
靠近AG3335MN的B8pin放置
靠近AG3335
NC in Tcxo
0R in Tcxo
Spare pad
800mA
GPIO4(SLOW SRC)-Internal pull low 8.33k
0:external (default)
1:internal (32k clock divided from DCXO26MHz clock
U2302
AG3335MN
U2301
BGU8019
FL2334
SAFFB1G56KB0F0AR15
Y2303
26MHz
Y2332_NM
0.032768 MHz
AG3335_GNSS_IN
GPS_L1_OUT
L1_LNA_EN
AG3335_OUTPUT_VIO18
AG3335_OUTPUT_VCORE
AG3335_OUTPUT_VRF
AVDD18_TCXO
AVDD18_AUXADC
AVSS_AUXADC
TSENSE_IN
OSC_XOUT
OSC_XIN
VREG_S4A_1P8
DC1V8_AG3335
GPS_32K
GPS_PWR_EN
AG3335_GPIO2_CTS
AG3335_GPIO3_RTS
AG3335_GPIO7_RX
AG3335_GPIO8_TX
AG3335MN_WAKE UP_CPU_INT
MCU_GPS_INT
Title
K81A_TEST_BOARD
Size
A3
Document Number
PM855B Charger
Rev
Date:
Thursday, May 27, 2021
Sheet



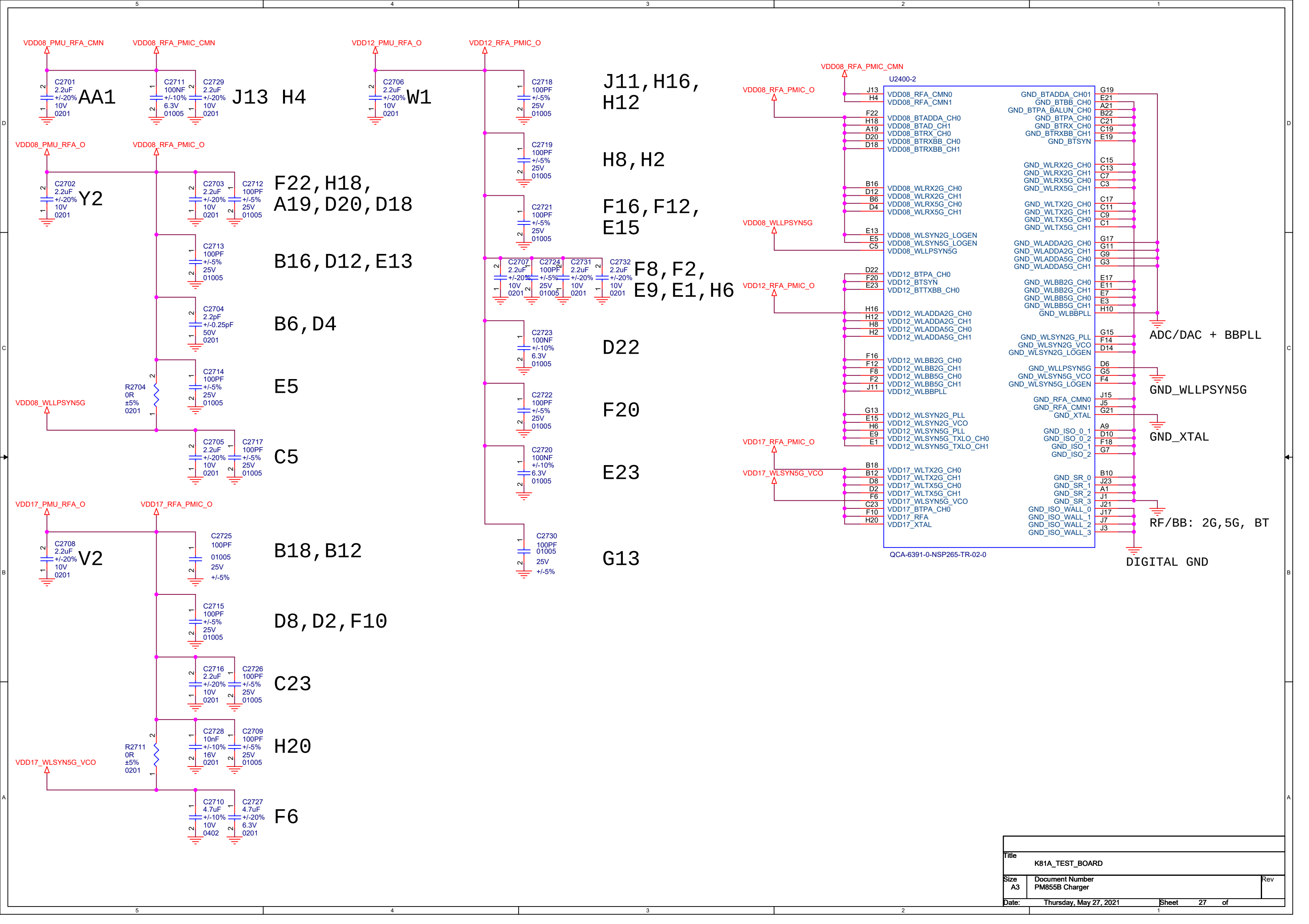
Title	K81A_TEST_BOARD		
Size A3	Document Number PM855B Charger		Rev
Date:	Thursday, May 27, 2021	Sheet 24	of



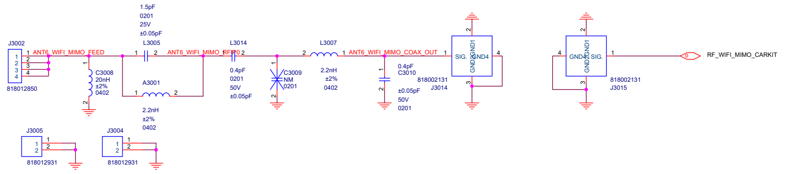
WLAN FEM 5G
WLAN FEM 2.4G
WLAN CH1
WLAN CH0
U2501
QM45391
U2500
QM42391
VPH_PWR
WL_RFRX_5G_CH0
WL_RFRX_5G_CH1
WL_RFTX_5G_CH0
WL_RFTX_5G_CH1
WLAN_5G_CH0
WLAN_5G_CH1
WL_RFPDET_5G_CH0
WL_RFPDET_5G_CH1
LNA_EN_5G_CH0
PA_EN_5G_CH0
BIAS_SEL_5G_CH0
LNA_EN_5G_CH1
PA_EN_5G_CH1
BIAS_SEL_5G_CH1
WLAN_2G_CH0
WLAN_2G_CH1
WL_RFRX_2G_CH0
WL_RFRX_2G_CH1
WL_RFTX_2G_CH0
WL_RFPDET_2G_CH0
WL_RFPDET_2G_CH1
LNA_EN_2G_CH0
PA_EN_2G_CH0
LNA_EN_2G_CH1
PA_EN_2G_CH1
BT_RFTX_CH0_FEM1
BT_RFTX_CH0_FEM0
WL_RFTX_2G_CH0
BT_EN_2G_CH0
BT_EN_2G_CH1
BT_RFTX_CH0_FEM0
BT_RFTX_CH0_FEM1
BT_RFTX_CH0
BT_XFEM_CTRL_ANT3
VREG_L16A_3P3
S2500
MXD8723E
RF_WIFI_MIMO_CARKIT
FL2500
TPX205950MT-7010A1
RF_WL5G_DIP_OUT_CH1
GPS_L1_OUT
RF_WL2G_DIP_OUT_CH1
RF_WIFI_CARKIT
FL2505
DPX105950DT-6012A7
RF_WL2G_TRIP_OUT_CH0
RF_WL5G_TRIP_OUT_CH0
靠近测试端口
靠近DPX105950DT-6012A7
RF_WL5G_DIP_OUT_CH1
RF_WL2G_DIP_OUT_CH1
RF_WL5G_TRIP_OUT_CH0
RF_WL2G_TRIP_OUT_CH0
WLAN_5G_CH1
WLAN_2G_CH1
WLAN_5G_CH0
WLAN_2G_CH0
FL2506_NM
1608-N5R5NAH
FL2504_NM
QM22545
FL2501_NM
1608-N5R5NAH
FL2502_NM
QM22545
Share pad
<Tolerance>
Title
K81A_TEST_BOARD
Size
A2
Document Number
PM855B Charger
Rev
Date:
Thursday, May 27, 2021
Sheet



Title	K81A_TEST_BOARD		
Size A3	Document Number PM855B Charger		Rev
Date:	Thursday, May 27, 2021	Sheet 26	of


VDD08_PMU_RFA_CMN
VDD08_RFA_PMIC_CMN
VDD12_PMU_RFA_O
VDD12_RFA_PMIC_O
C2701 2.2uF +/-20% 10V 0201
AA1
C2711 100NF +/-10% 6.3V 01005
C2729 2.2uF +/-20% 10V 0201
J13 H4
C2706 2.2uF +/-20% 10V 0201
W1
C2718 100PF +/-5% 25V 01005
J11,H16,
H12
VDD08_PMU_RFA_O
VDD08_RFA_PMIC_O
C2702 2.2uF +/-20% 10V 0201
Y2
C2703 2.2uF +/-20% 10V 0201
C2712 100PF +/-5% 25V 01005
F22,H18,
A19,D20,D18
C2719 100PF +/-5% 25V 01005
H8,H2
C2721 100PF +/-5% 25V 01005
F16,F12,
E15
C2713 100PF +/-5% 25V 01005
B16,D12,E13
C2707 2.2uF +/-20% 10V 0201
C2724 100PF +/-5% 25V 01005
C2731 2.2uF +/-20% 10V 0201
C2732 2.2uF +/-20% 10V 0201
F8,F2,
E9,E1,H6
C2704 2.2pF +/-0.25pF 50V 0201
B6,D4
C2723 100NF +/-10% 6.3V 01005
D22
C2714 100PF +/-5% 25V 01005
E5
R2704 0R ±5% 0201
VDD08_WLLPSYN5G
C2722 100PF +/-5% 25V 01005
F20
C2705 2.2uF +/-20% 10V 0201
C2717 100PF +/-5% 25V 01005
C5
C2720 100NF +/-10% 6.3V 01005
E23
VDD17_PMU_RFA_O
VDD17_RFA_PMIC_O
C2708 2.2uF +/-20% 10V 0201
V2
C2725 100PF 01005 25V +/-5%
B18,B12
C2730 100PF 01005 25V +/-5%
G13
C2715 100PF +/-5% 25V 01005
D8,D2,F10
C2716 2.2uF +/-20% 10V 0201
C2726 100PF +/-5% 25V 01005
C23
C2728 10nF +/-10% 16V 0201
C2709 100PF +/-5% 25V 01005
H20
R2711 0R ±5% 0201
VDD17_WLSYN5G_VCO
C2710 4.7uF +/-10% 10V 0402
C2727 4.7uF +/-20% 6.3V 0201
F6
VDD08_RFA_PMIC_CMN
U2400-2
VDD08_RFA_PMIC_O
VDD08_WLLPSYN5G
VDD12_RFA_PMIC_O
VDD17_RFA_PMIC_O
VDD17_WLSYN5G_VCO
J13 VDD08_RFA_CMN0
H4 VDD08_RFA_CMN1
F22 VDD08_BTADDA_CH0
H18 VDD08_BTAD_CH1
A19 VDD08_BTRX_CH0
D20 VDD08_BTRXBB_CH0
D18 VDD08_BTRXBB_CH1
B16 VDD08_WLRX2G_CH0
D12 VDD08_WLRX2G_CH1
B6 VDD08_WLRX5G_CH0
D4 VDD08_WLRX5G_CH1
E13 VDD08_WLSYN2G_LOGEN
E5 VDD08_WLSYN5G_LOGEN
C5 VDD08_WLLPSYN5G
D22 VDD12_BTPA_CH0
F20 VDD12_BTSYN
E23 VDD12_BTTXBB_CH0
H16 VDD12_WLADDA2G_CH0
H12 VDD12_WLADDA2G_CH1
H8 VDD12_WLADDA5G_CH0
H2 VDD12_WLADDA5G_CH1
F16 VDD12_WLBB2G_CH0
F12 VDD12_WLBB2G_CH1
F8 VDD12_WLBB5G_CH0
F2 VDD12_WLBB5G_CH1
J11 VDD12_WLBBPLL
G13 VDD12_WLSYN2G_PLL
E15 VDD12_WLSYN2G_VCO
H6 VDD12_WLSYN5G_PLL
E9 VDD12_WLSYN5G_TXLO_CH0
E1 VDD12_WLSYN5G_TXLO_CH1
B18 VDD17_WLTX2G_CH0
B12 VDD17_WLTX2G_CH1
D8 VDD17_WLTX5G_CH0
D2 VDD17_WLTX5G_CH1
F6 VDD17_WLSYN5G_VCO
C23 VDD17_BTPA_CH0
F10 VDD17_RFA
H20 VDD17_XTAL
GND_BTADDA_CH01 G19
GND_BTBB_CH0 E21
GND_BTPA_BALUN_CH0 A21
GND_BTPA_CH0 B22
GND_BTRX_CH0 C21
GND_BTRXBB_CH1 C19
GND_BTSYN E19
GND_WLRX2G_CH0 C15
GND_WLRX2G_CH1 C13
GND_WLRX5G_CH0 C7
GND_WLRX5G_CH1 C3
GND_WLTX2G_CH0 C17
GND_WLTX2G_CH1 C11
GND_WLTX5G_CH0 C9
GND_WLTX5G_CH1 C1
GND_WLADDA2G_CH0 G17
GND_WLADDA2G_CH1 G11
GND_WLADDA5G_CH0 G9
GND_WLADDA5G_CH1 G3
GND_WLBB2G_CH0 E17
GND_WLBB2G_CH1 E11
GND_WLBB5G_CH0 E7
GND_WLBB5G_CH1 E3
GND_WLBBPLL H10
GND_WLSYN2G_PLL G15
GND_WLSYN2G_VCO F14
GND_WLSYN2G_LOGEN D14
GND_WLLPSYN5G D6
GND_WLSYN5G_VCO G5
GND_WLSYN5G_LOGEN F4
GND_RFA_CMN0 J15
GND_RFA_CMN1 J5
GND_XTAL G21
GND_ISO_0_1 A9
GND_ISO_0_2 D10
GND_ISO_1 F18
GND_ISO_2 G7
GND_SR_0 B10
GND_SR_1 J23
GND_SR_2 A1
GND_SR_3 J1
GND_ISO_WALL_0 J21
GND_ISO_WALL_1 J17
GND_ISO_WALL_2 J7
GND_ISO_WALL_3 J3
QCA-6391-0-NSP265-TR-02-0
ADC/DAC + BBPLL
GND_WLLPSYN5G
GND_XTAL
RF/BB: 2G,5G, BT
DIGITAL GND
Title K81A_TEST_BOARD
Size A3
Document Number PM855B Charger
Rev
Date: Thursday, May 27, 2021
Sheet 27 of

ANT6 WIFI1



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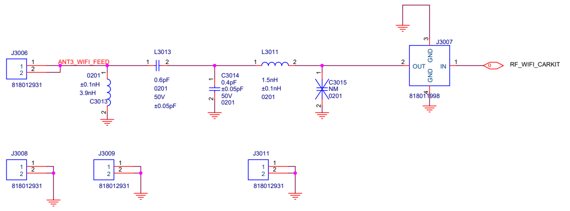
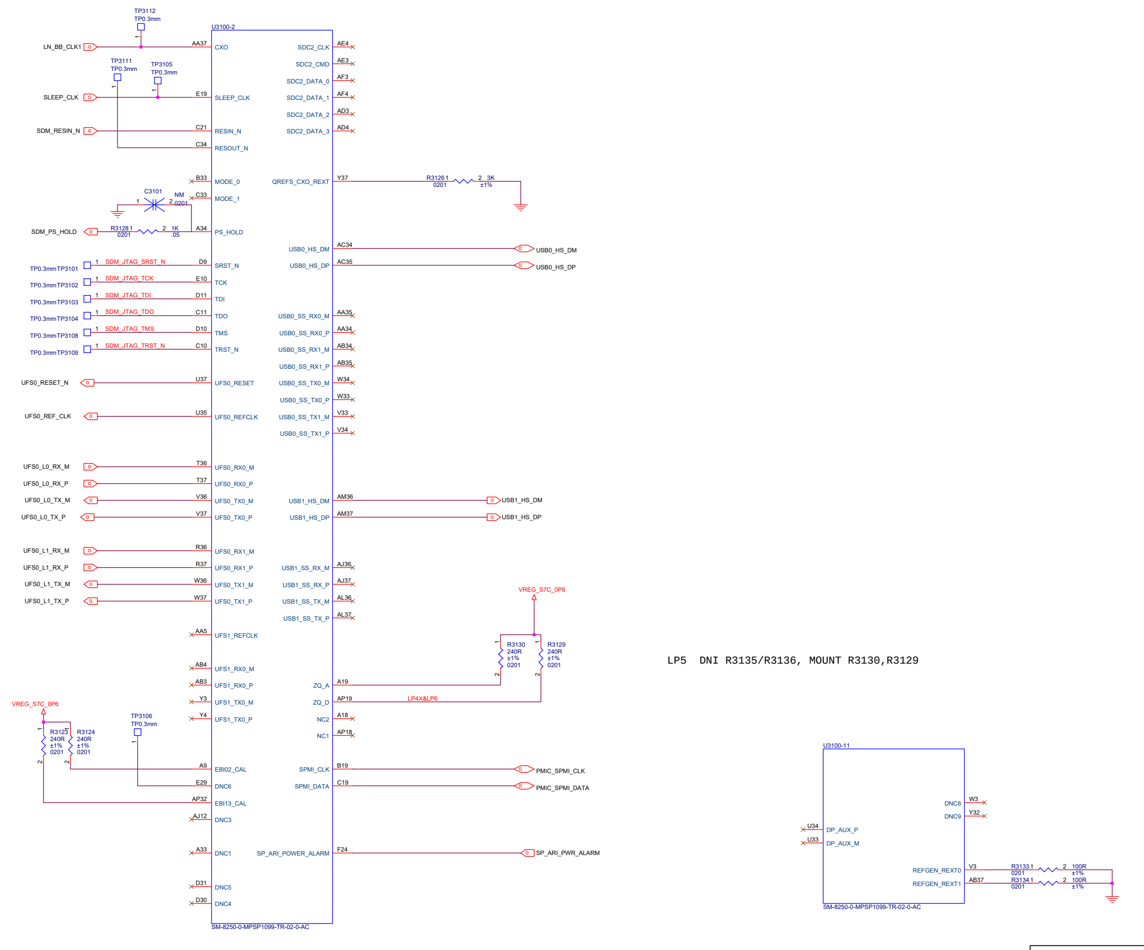


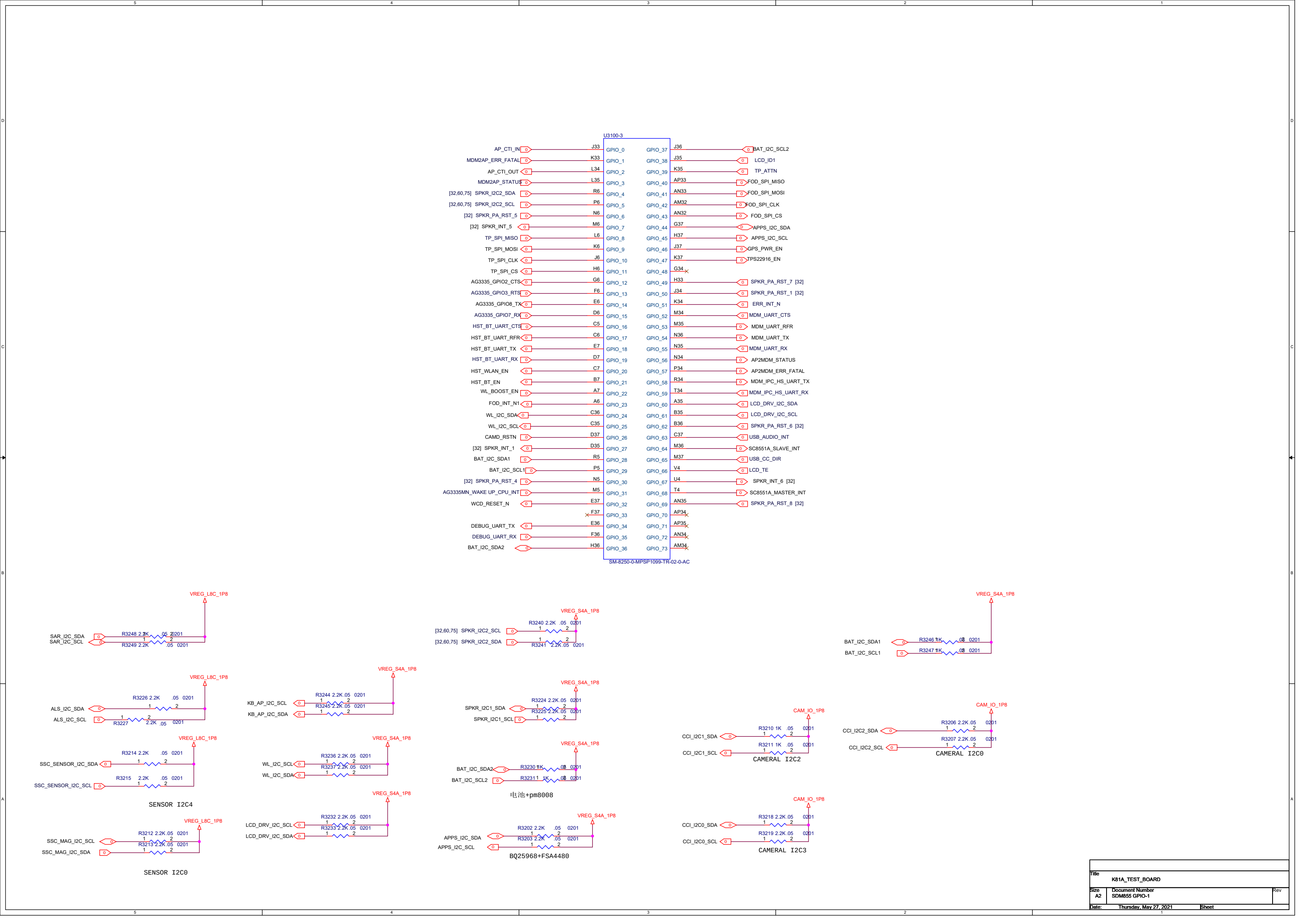
Title	K81A_TEST_BOARD
Size A2	Document Number PM855B Charger
Rev	
Date:	Thursday, May 27, 2021
Sheet	

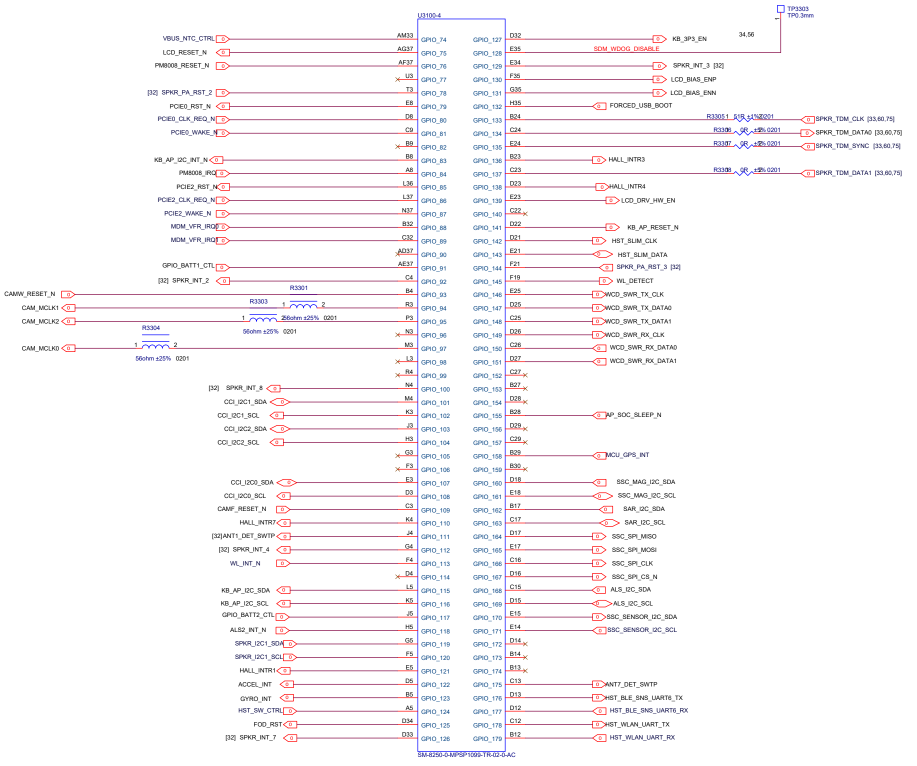
We use current SM8250 symbol as LP5 PKG, and it can be compatible with LP4x PKG



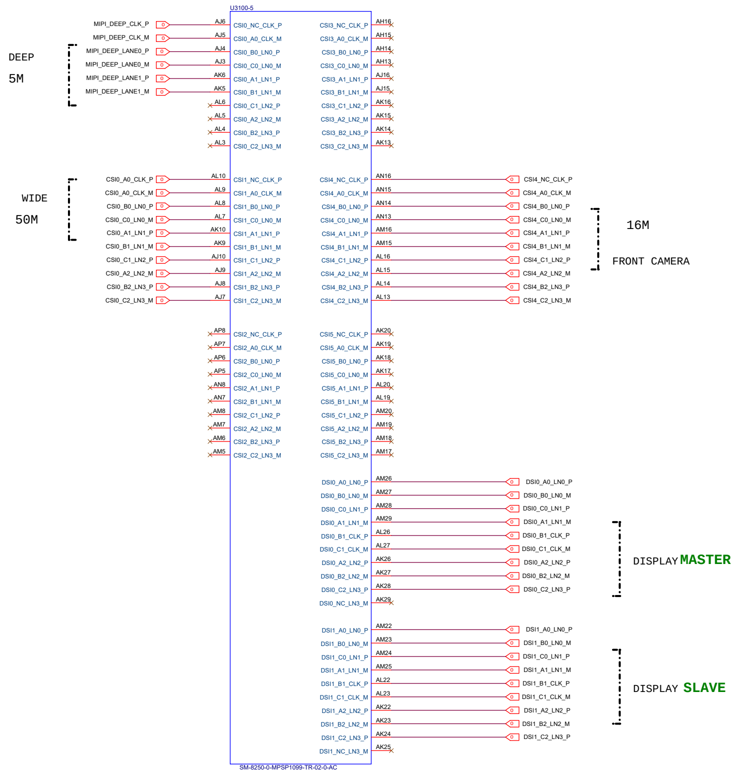
LP5 DNI R3135/R3136, MOUNT R3130,R3129

Title	K81A_TEST_BOARD	
Size A2	Document Number SDM855 Control	Rev
Date:	Thursday, May 27, 2021	Sheet


U3100-3
SM-8250-0-MPSP1099-TR-02-0-AC
SENSOR I2C4
SENSOR I2C0
CAMERAL I2C2
CAMERAL I2C0
CAMERAL I2C3
电池+pm8008
BQ25968+FSA4480
VREG_L8C_1P8
VREG_S4A_1P8
CAM_IO_1P8
K81A_TEST_BOARD
SDM855 GPIO-1
Thursday, May 27, 2021

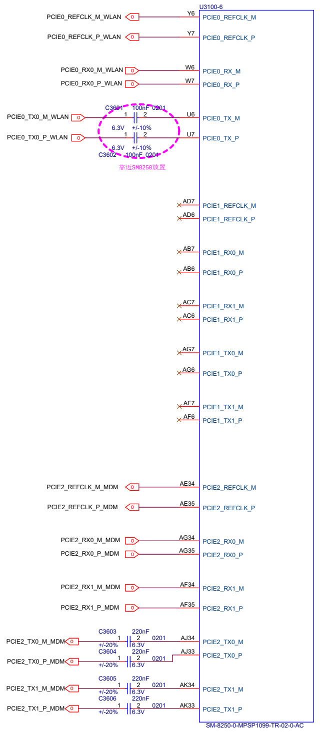


Title	K81A_TEST_BOARD	
Size A2	Document Number SDM855 GPIO-2	Rev
Date: Thursday, May 27, 2021	Sheet	



Title	K81A_TEST_BOARD	
Size A2	Document Number SDM855 MIPI	Rev
Date: Thursday, May 27, 2021	Sheet	

Title	K81A_TEST_BOARD	
Size A3	Document Number Reserved(for MIPI)	Rev
Date: Thursday, May 27, 2021	Sheet 36 of	



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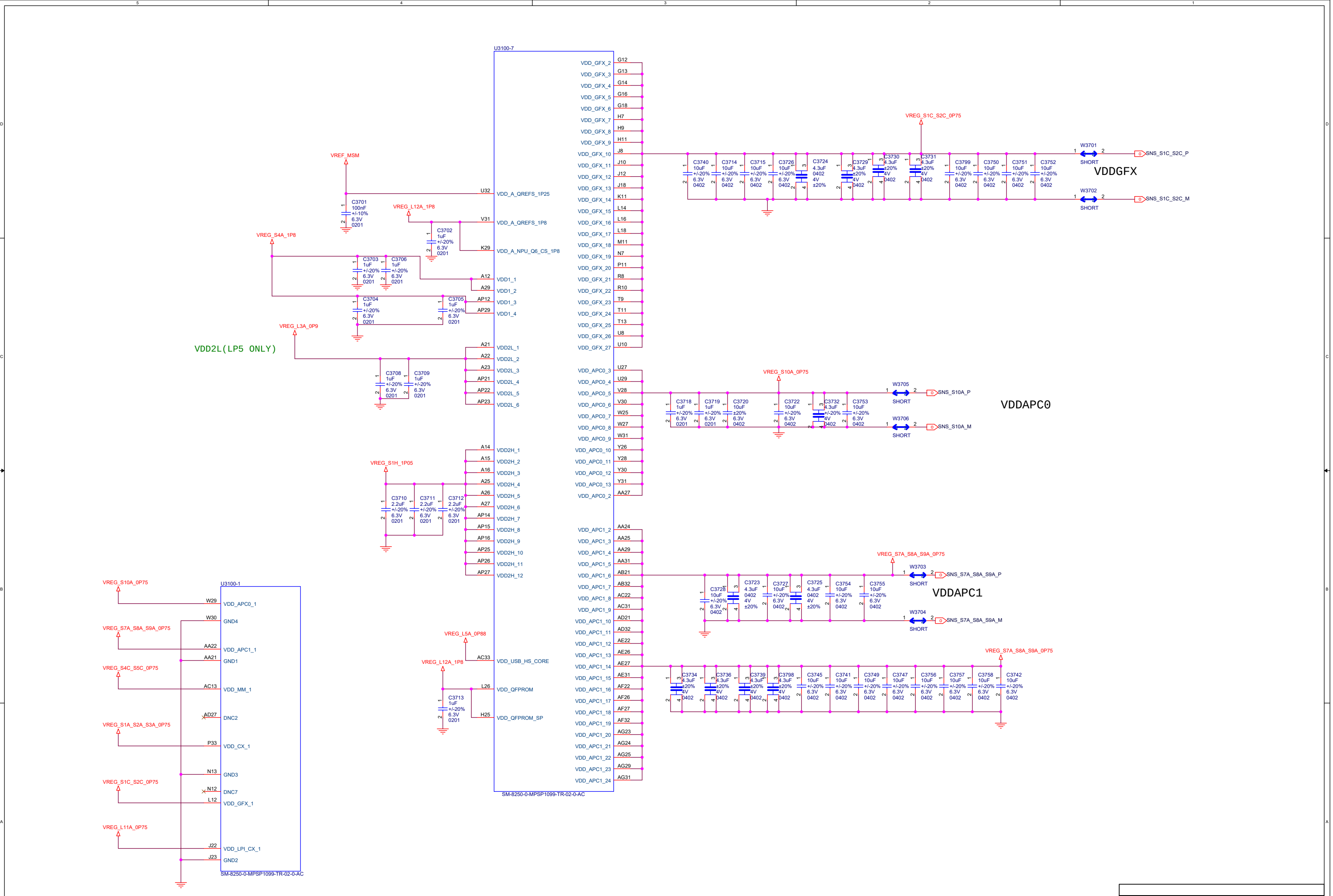
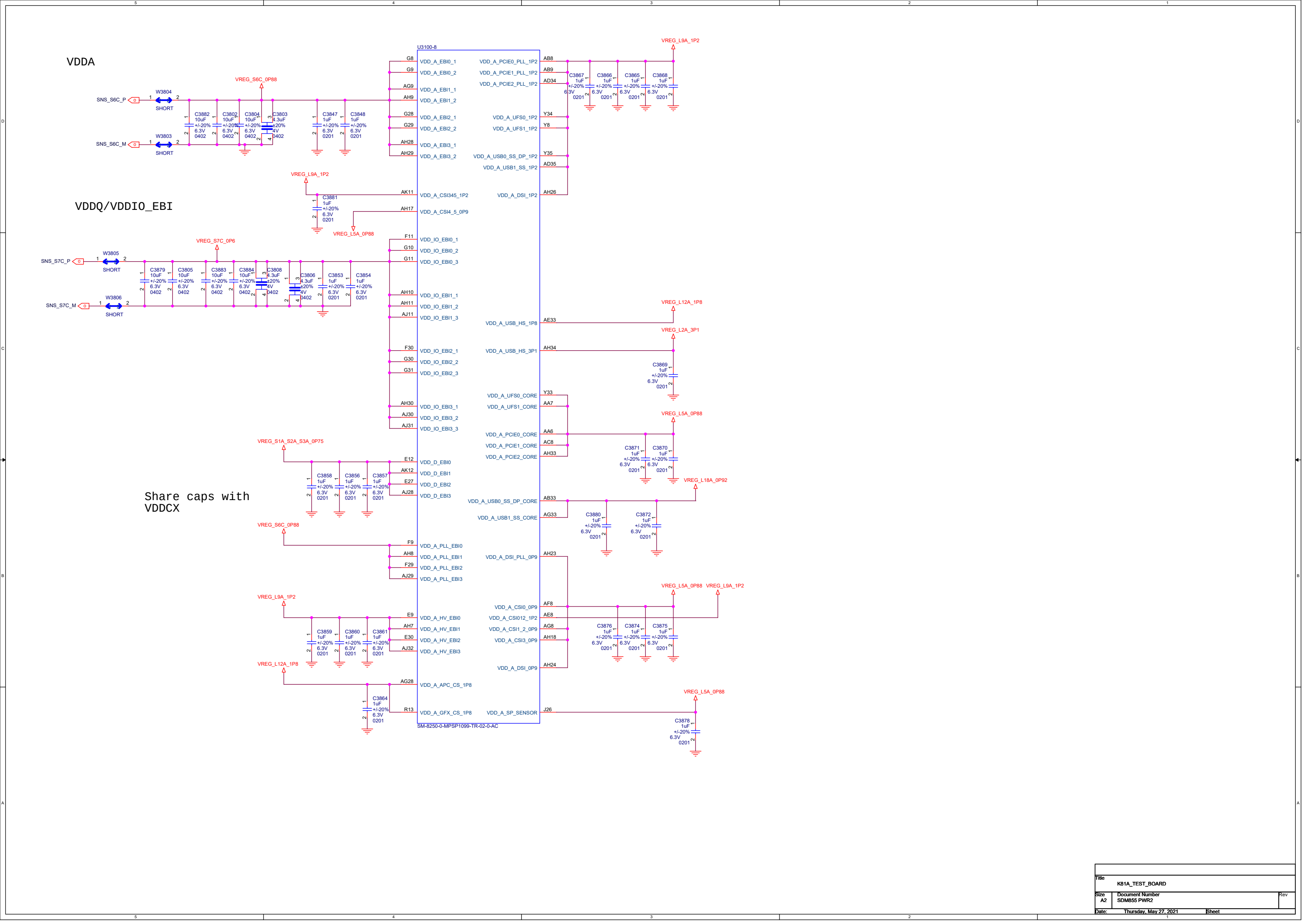
Size A2	Document Number SDM855 RF INTF	Rev
Date: Thursday, May 27, 2021	Sheet	

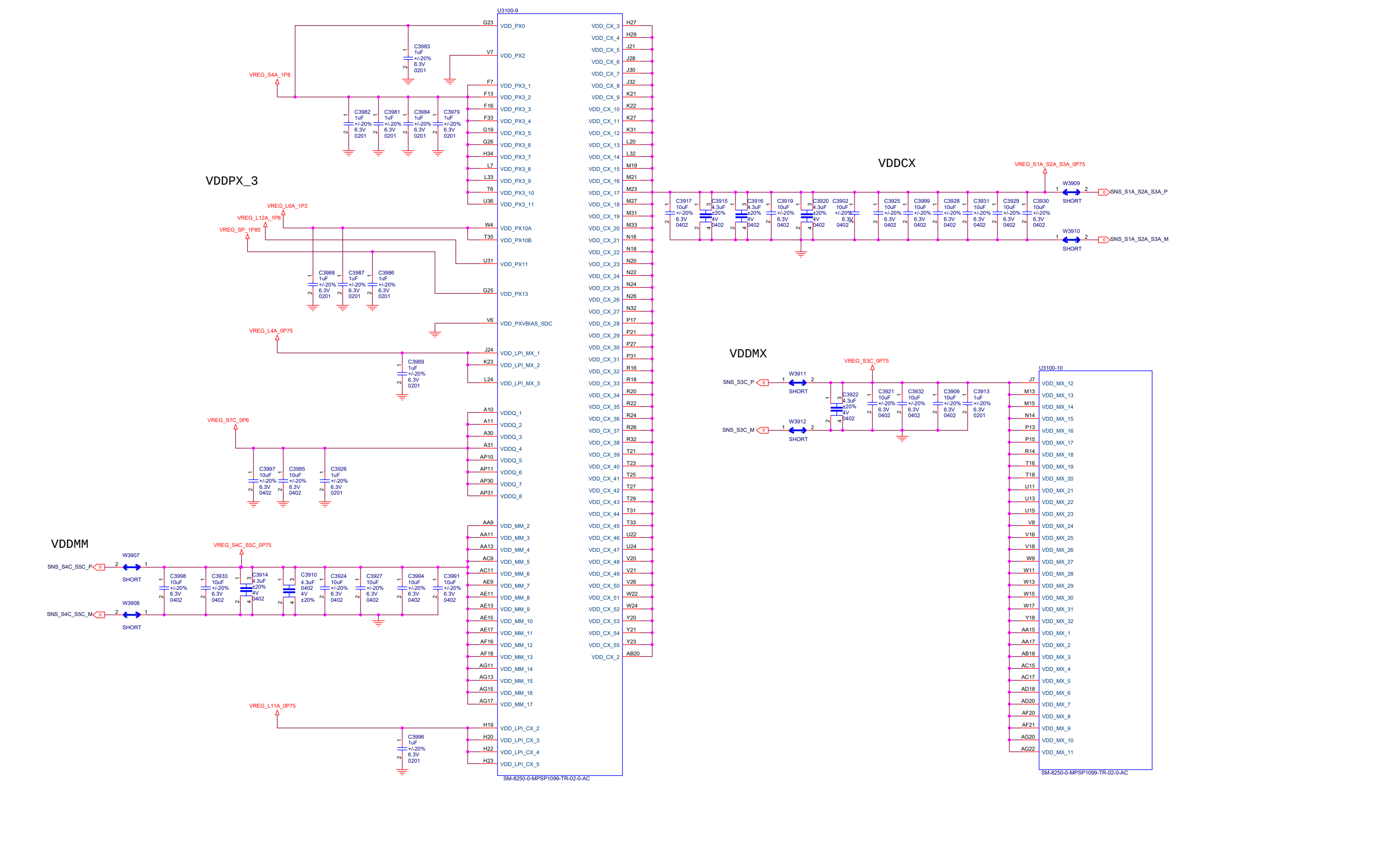


Title
K81A_TEST_BOARD

Size	Document Number	Rev
A2	SDM855 PWR1	

Date: Thursday, May 27, 2021 Sheet


VDDA
VDDQ/VDDIO_EBI
Share caps with VDDCX
U3100-8
SM-8250-0-MPSP1099-TR-02-0-AC
VDD_A_EBI0_1
VDD_A_EBI0_2
VDD_A_EBI1_1
VDD_A_EBI1_2
VDD_A_EBI2_1
VDD_A_EBI2_2
VDD_A_EBI3_1
VDD_A_EBI3_2
VDD_A_CSI345_1P2
VDD_A_CSI4_5_0P9
VDD_IO_EBI0_1
VDD_IO_EBI0_2
VDD_IO_EBI0_3
VDD_IO_EBI1_1
VDD_IO_EBI1_2
VDD_IO_EBI1_3
VDD_IO_EBI2_1
VDD_IO_EBI2_2
VDD_IO_EBI2_3
VDD_IO_EBI3_1
VDD_IO_EBI3_2
VDD_IO_EBI3_3
VDD_D_EBI0
VDD_D_EBI1
VDD_D_EBI2
VDD_D_EBI3
VDD_A_PLL_EBI0
VDD_A_PLL_EBI1
VDD_A_PLL_EBI2
VDD_A_PLL_EBI3
VDD_A_HV_EBI0
VDD_A_HV_EBI1
VDD_A_HV_EBI2
VDD_A_HV_EBI3
VDD_A_APC_CS_1P8
VDD_A_GFX_CS_1P8
VDD_A_PCIE0_PLL_1P2
VDD_A_PCIE1_PLL_1P2
VDD_A_PCIE2_PLL_1P2
VDD_A_UFS0_1P2
VDD_A_UFS1_1P2
VDD_A_USB0_SS_DP_1P2
VDD_A_USB1_SS_1P2
VDD_A_DSI_1P2
VDD_A_USB_HS_1P8
VDD_A_USB_HS_3P1
VDD_A_UFS0_CORE
VDD_A_UFS1_CORE
VDD_A_PCIE0_CORE
VDD_A_PCIE1_CORE
VDD_A_PCIE2_CORE
VDD_A_USB0_SS_DP_CORE
VDD_A_USB1_SS_CORE
VDD_A_DSI_PLL_0P9
VDD_A_CSI0_0P9
VDD_A_CSI012_1P2
VDD_A_CSI1_2_0P9
VDD_A_CSI3_0P9
VDD_A_DSI_0P9
VDD_A_SP_SENSOR
SNS_S6C_P
SNS_S6C_M
SNS_S7C_P
SNS_S7C_M
W3803
W3804
W3805
W3806
SHORT
VREG_S6C_0P88
VREG_L9A_1P2
VREG_L5A_0P88
VREG_S7C_0P6
VREG_S1A_S2A_S3A_0P75
VREG_L12A_1P8
VREG_L2A_3P1
VREG_L18A_0P92
Title
K81A_TEST_BOARD
Size
A2
Document Number
SDM855 PWR2
Rev
Date:
Thursday, May 27, 2021
Sheet

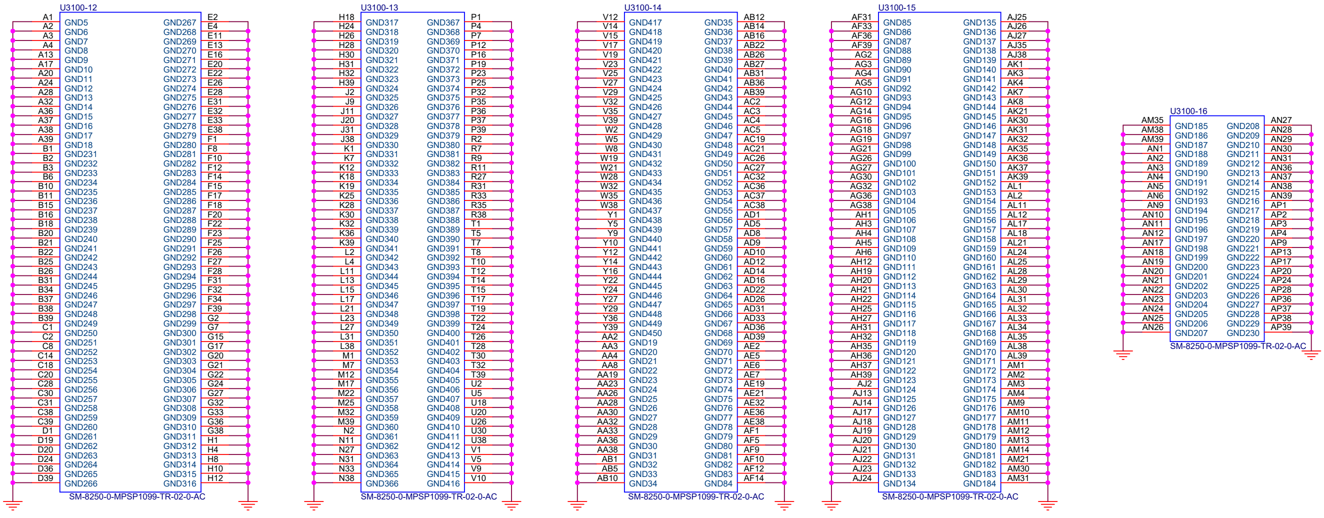


Title
K81A_TEST_BOARD

Size	Document Number	Rev
A2	SDM855 PWR3	

Date: Thursday, May 27, 2021 Sheet

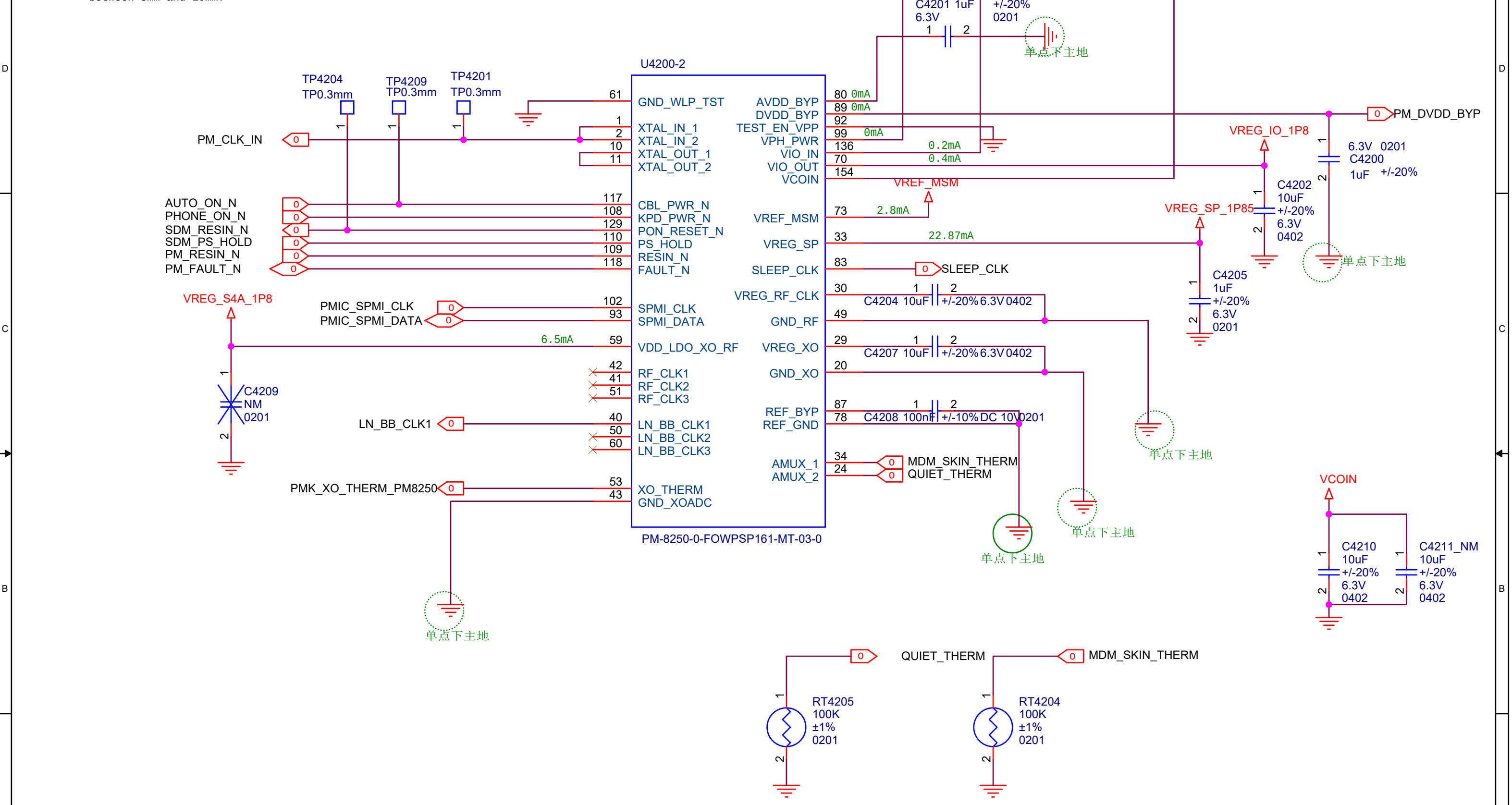
Title	K81A_TEST_BOARD	
Size A3	Document Number SDM855 PWR4	Rev
Date: Thursday, May 27, 2021	Sheet 41 of	



Title	K81A_TEST_BOARD		
Size A3	Document Number SDM855 GND		Rev
Date:	Thursday, May 27, 2021	Sheet 42	of

Note: Refer to 80-VP447-10 for XO Layout App Note.

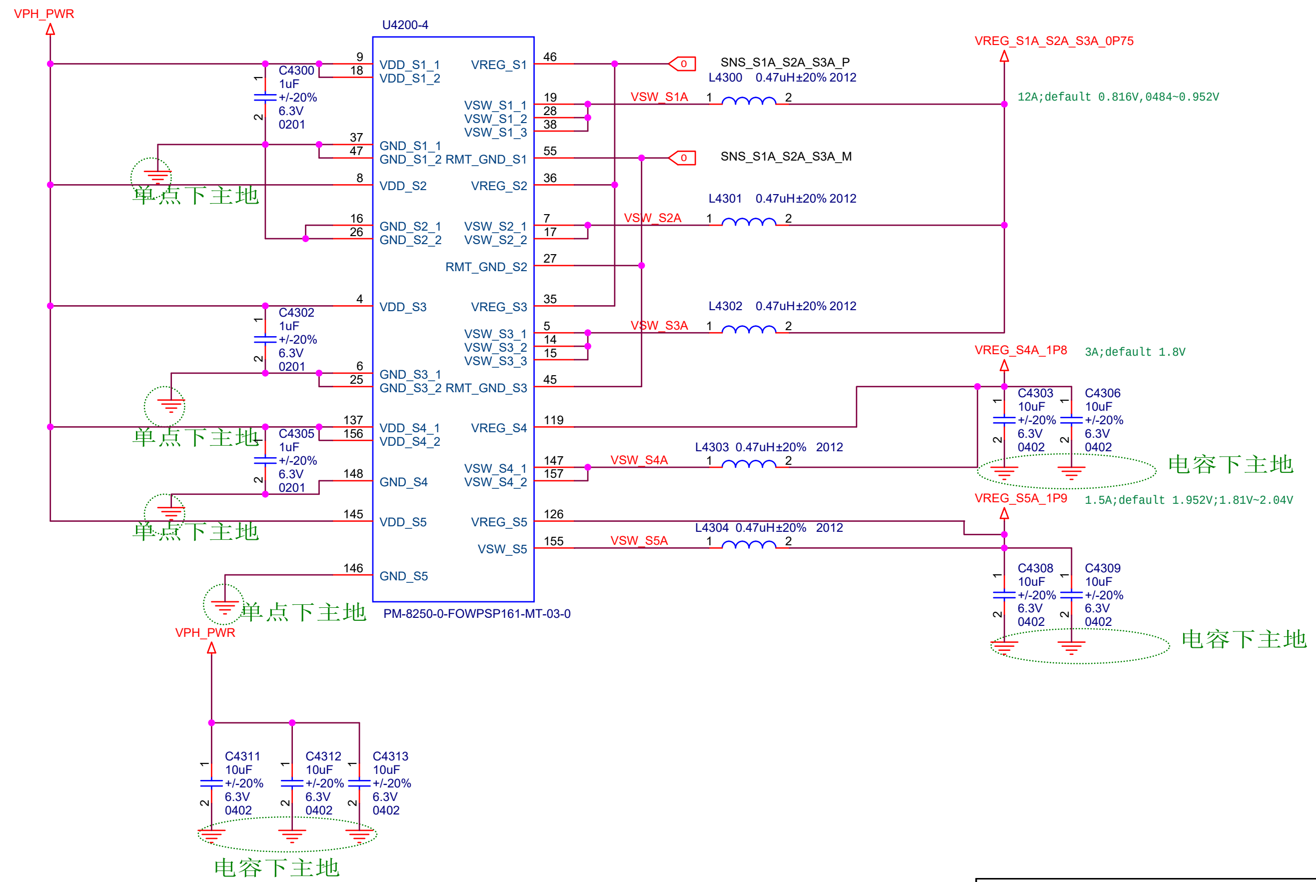
Layout Note:
No trace shall be present in at-least two immediate layers below XTAL.
Route XTAL_IN and XTAL_OUT traces on surface layer only, with length between 3mm and 10mm.



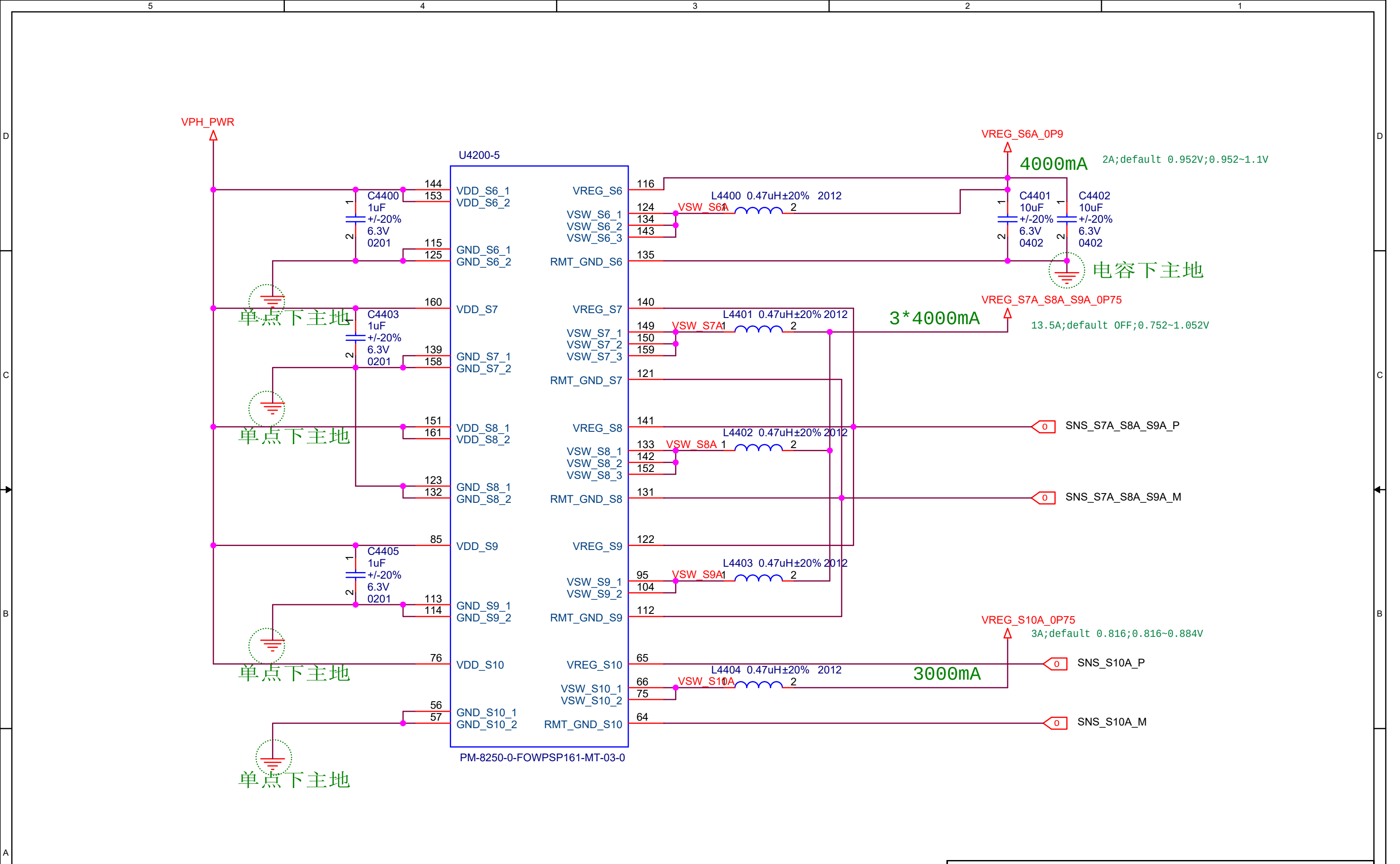
PLACE R4204 TO A COOL PLACE

Close to MSM8250

Title	K81A_TEST_BOARD
Size A4	Document Number PM855 XO&CTRL
Rev	
Date:	Thursday, May 27, 2021
Sheet	43



Title	K81A_TEST_BOARD		
Size A4	Document Number PM855 BUCK CONVERTER1		Rev
Date:	Thursday, May 27, 2021	Sheet	44



Title	K81A_TEST_BOARD	
Size A4	Document Number PM855 BUCK CONVERTER2	Rev
Date: Thursday, May 27, 2021	Sheet 45	

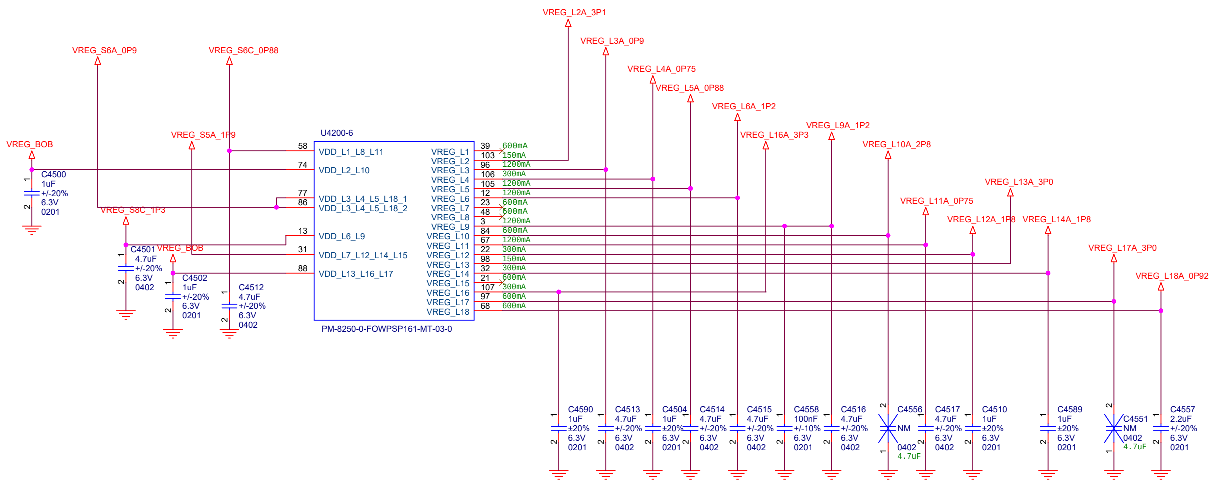


Table 3-9 PM8250 linear/low-voltage regulator summary

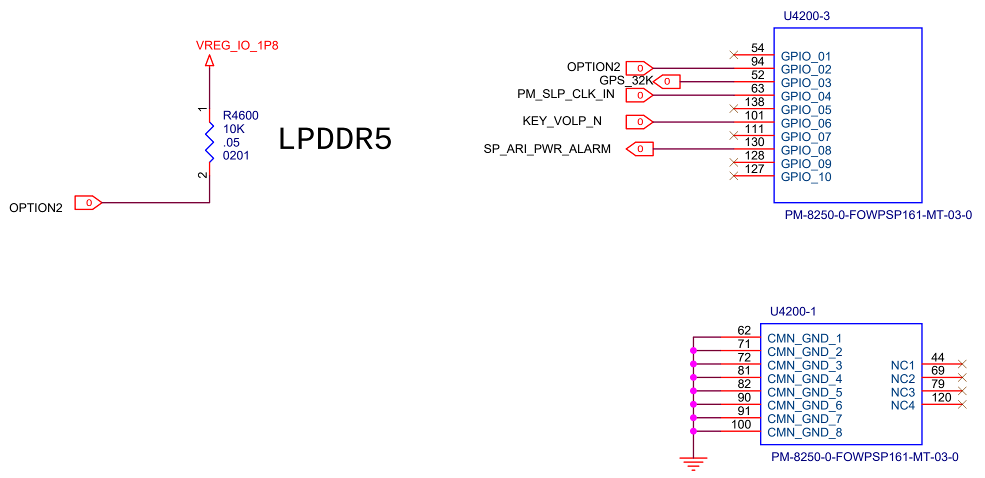
Regulator	Circuit type	I_{RATED} (mA) [1]	Default on booting voltage (V) [2]	Nominal voltage (V)	Active voltage minimum (V)	Active voltage maximum (V)	Expected use
L1A	LDO510 NMOS 600	600	–	0.752			Spare
L2A	LDO510 MV PMOS 150	150	3.072	3.072	3.072	3.072	USB
L3A	LDO510 NMOS 1200	1200	0.928	0.928			LPDDR5: VDD2L LPDDR4X: Spare
L4A	LDO510 NMOS 300	300	0.816	0.816	0.752	0.904	LPI_MX
L5A	LDO510 NMOS 1200	1200	0.88	0.88	0.88	0.88	LPI_MX
L6A	LDO510 NMOS 1200	1200	1.2	1.2	1.2	1.2	UFS VCCQ, PX_10
L7A	LDO510 LV PMOS 600	600	–	1.704			QCA6390 VDDRFA backup
L8A	LDO510 NMOS 600	600	–	0.816			Spare
L9A	LDO510 NMOS 1200	1200	1.2	1.2	1.2	1.2	1.2V VDDA for DSI, PCIE, HV EBI, USB_SS_DP_1P2, CSI, and so on.

Table 3-9 PM8250 linear/low-voltage regulator summary (cont.)

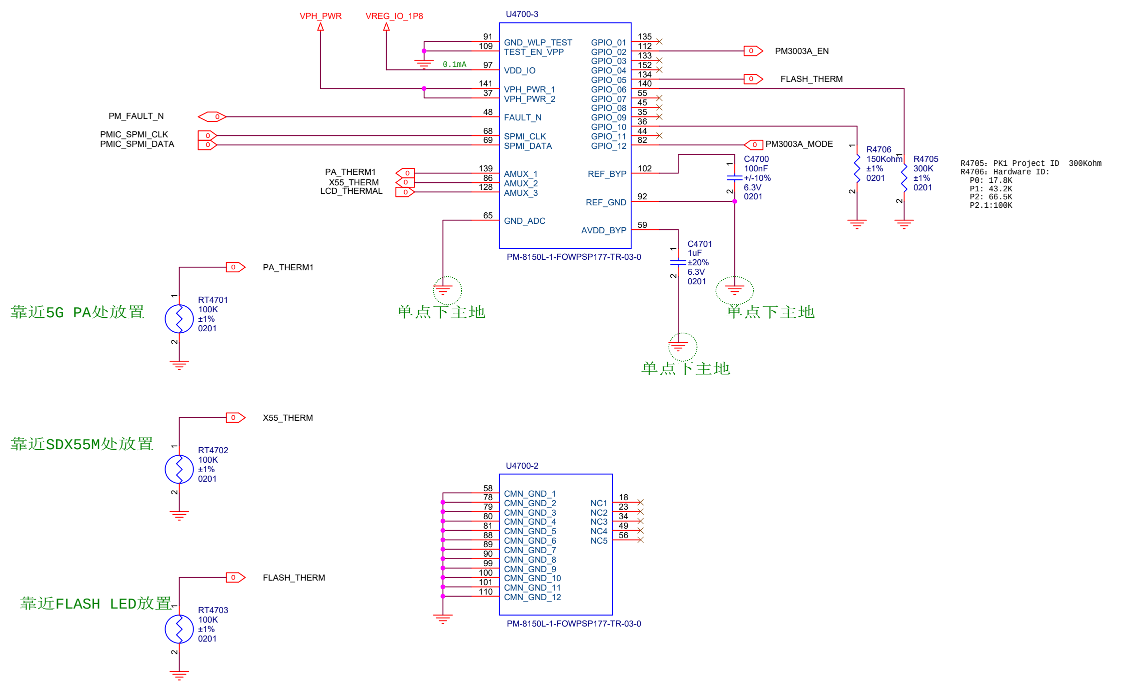
Regulator	Circuit type	I_{RATED} (mA) [1]	Default on booting voltage (V) [2]	Nominal voltage (V)	Active voltage minimum (V)	Active voltage maximum (V)	Expected use
L10A	LDO510 MV PMOS 600	600	–	2.904			Optional DMIC Bias
L11A	LDO510 NMOS 1200	1200	0.816	0.816			LPI_CX
L12A	LDO510 LV P300	300	1.8	1.8	1.8	1.8	BBCLK, USB, UFS, QFPROM
L13A	LDO510 MV P150	150	–	3.008			FTM TSP_AVDD
L14A	LDO510 LV P300	300	–	1.8	1.8	1.8	OLED_VDDIO
L15A	LDO510 LV P600	600	–	1.704	1.704	1.704	QCA6436 VCC_1P8
L16A	LDO510 MV P300	300	–	3.024			Ext BT xPA 3p3
L17A	LDO510 MV P600	600 [3]	2.96	2.96	2.856	3.008	UFS_2L_VCC
L18A	LDO510 N600	600	0.912	0.912			USB SS core & DP core

1. Some regulators are tested and used beyond their lrated limit
2. Voltages shown in the *Default on booting voltage (V)* column indicate that the regulator is enabled as part of the hardware default-on sequence.
3. VREG_L17A has been specifically validated up to 800 mA for UFS VSS support.

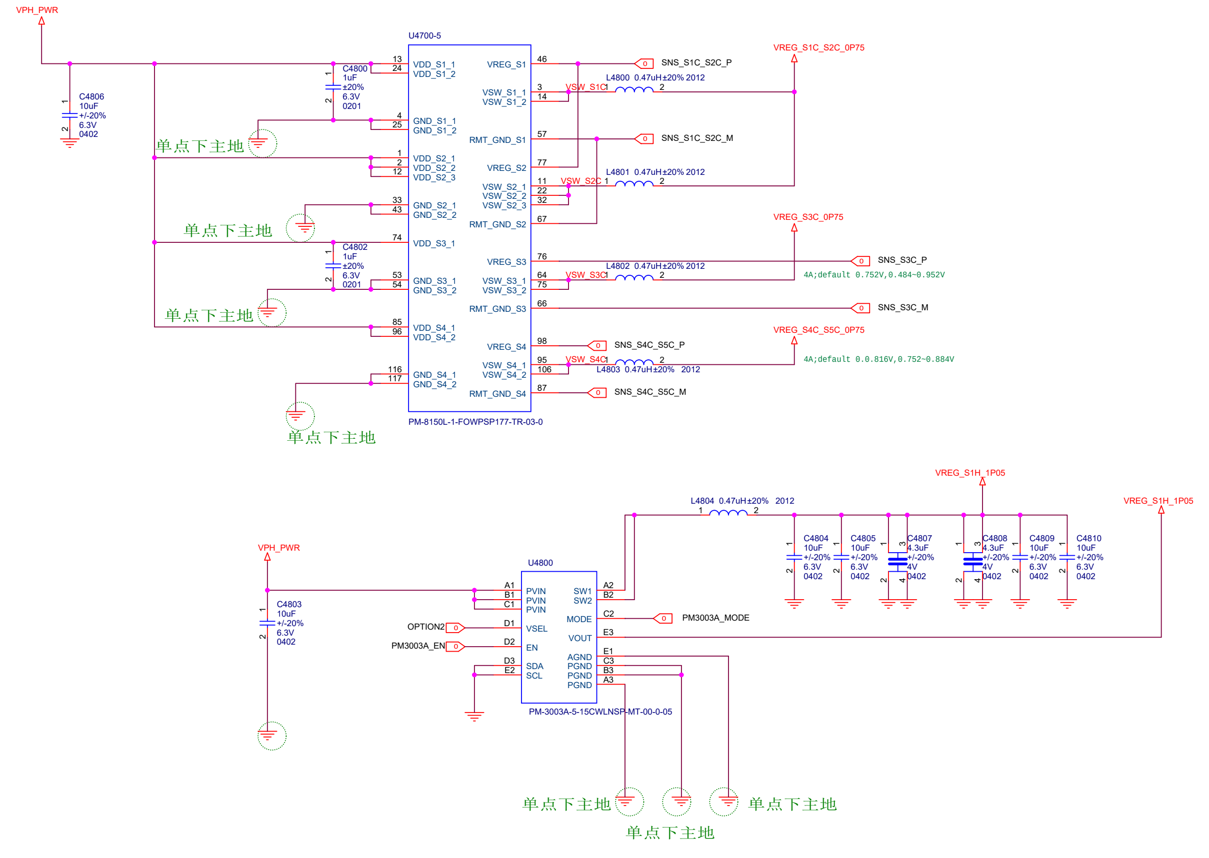
Title	K81A_TEST_BOARD
Size A3	Document Number PM855 LDOs
Rev	
Date:	Thursday, May 27, 2021
Sheet	46 of 77



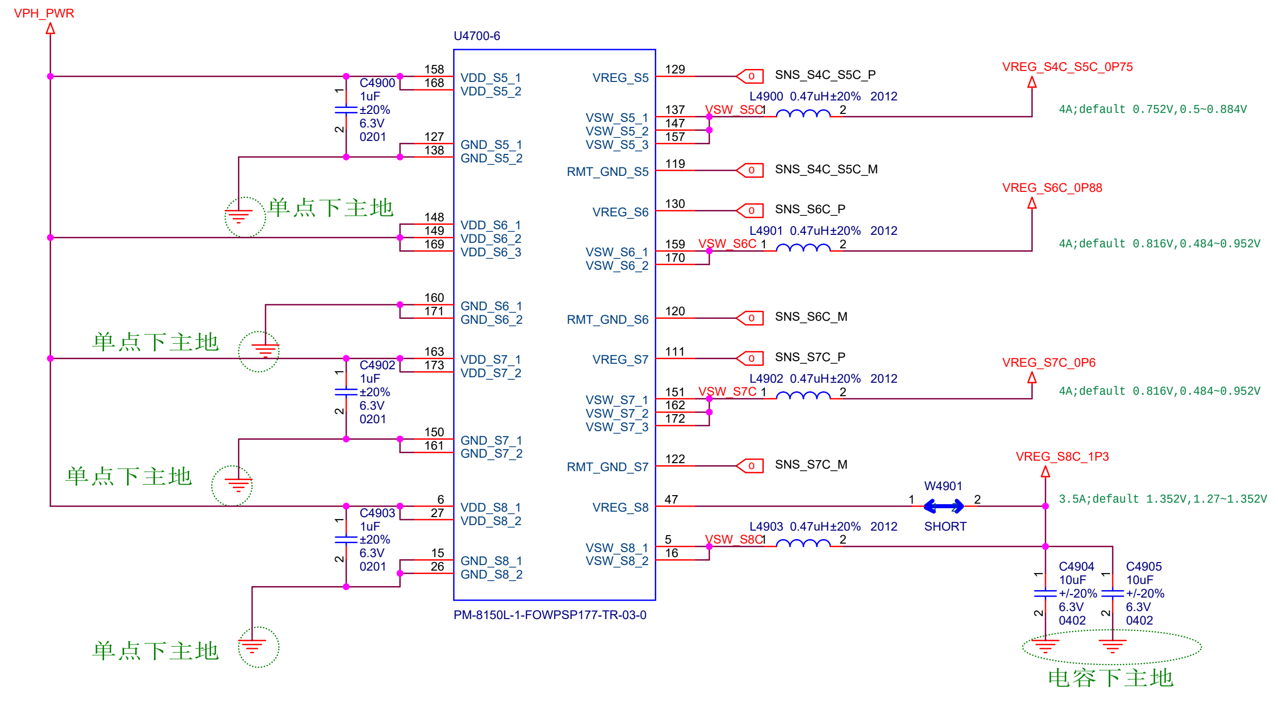
Title	K81A_TEST_BOARD		
Size A4	Document Number PM855 GPIO&GND		Rev
Date:	Thursday, May 27, 2021	Sheet	47



Title	K81A_TEST_BOARD
Size A3	Document Number PM855A CTRL&GND
Date: Friday, May 28, 2021	Sheet 48 of



Title	K81A_TEST_BOARD	
Size A3	Document Number PM855A BUCK CONVERTER1	Rev
Date: Thursday, May 27, 2021	Sheet 49 of 77	

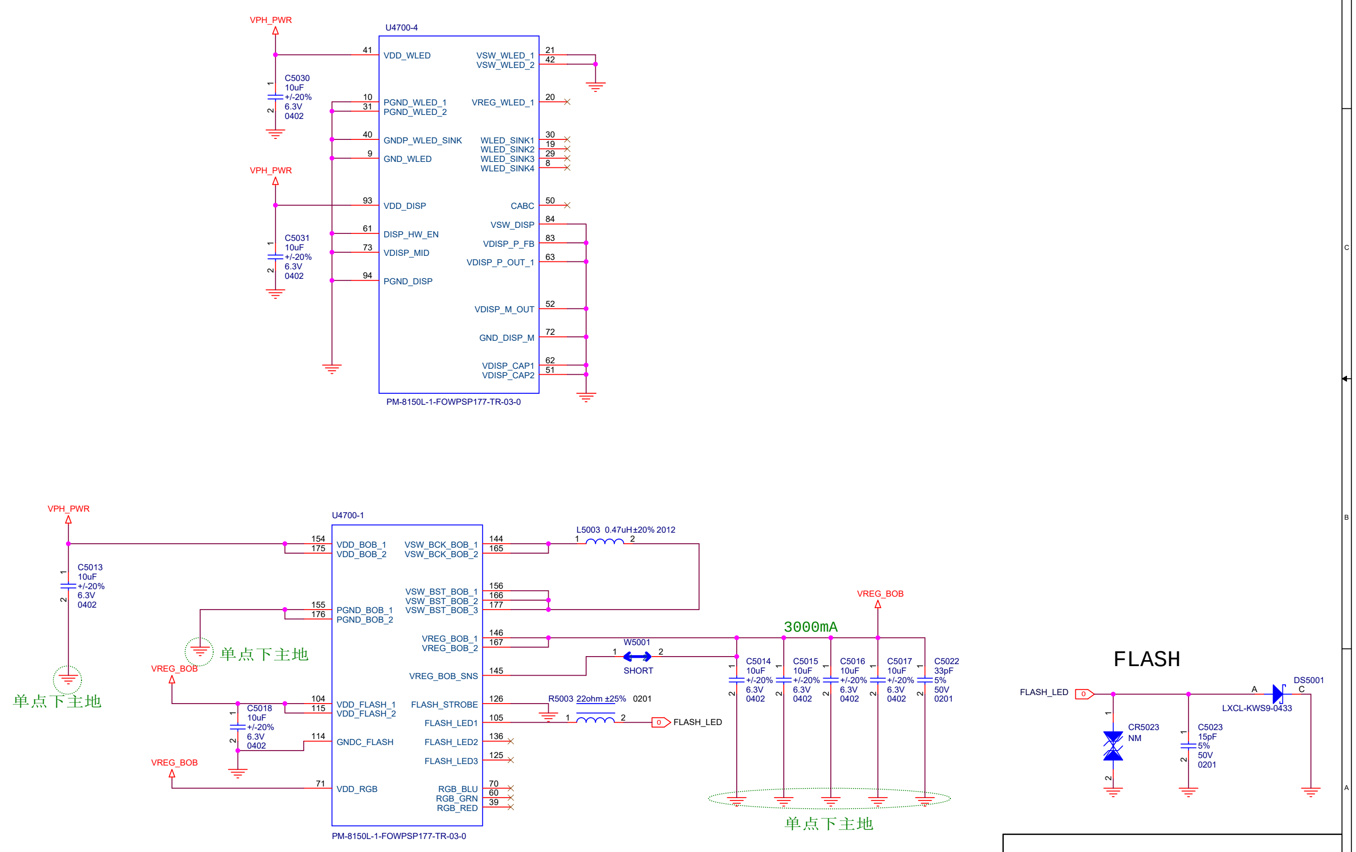


Title: K81A_TEST_BOARD

Size	Document Number	Rev
A4	PM855A BUCK CONVERTER2	

Date: Thursday, May 27, 2021 Sheet 49

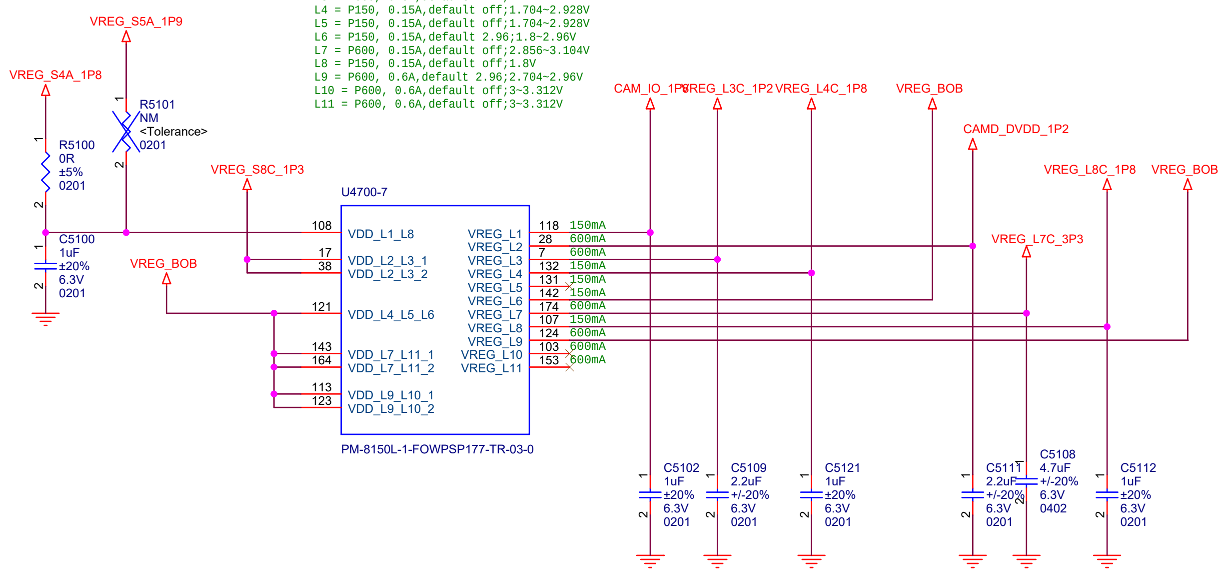
Make sure SCP(short circuit peotection) must be disabled



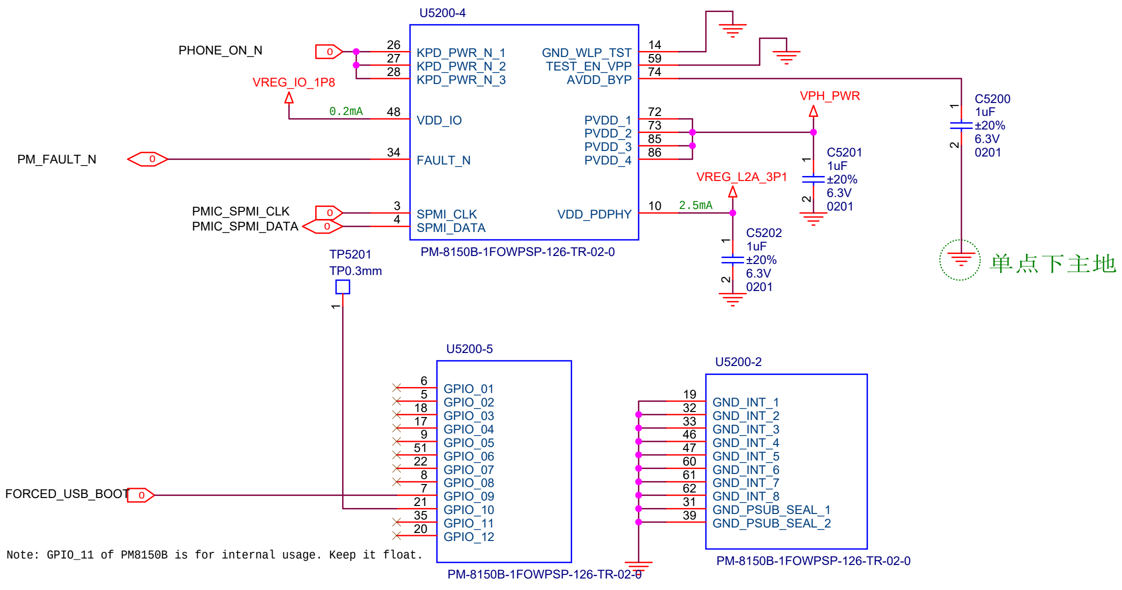
Title	K81A_TEST_BOARD	
Size A3	Document Number PM855A AMOLED&BOB&FLASH	Rev
Date: Thursday, May 27, 2021	Sheet 50 of	

NMOS and LV PMOS ;HEADROOM=100mV

L1 = P150, 0.15A,default off;1.8V
L2 = N600, 0.15A,default off;1.304V
L3 = N600, 0.6A,default 1.2V;1.2V
L4 = P150, 0.15A,default off;1.704~2.928V
L5 = P150, 0.15A,default off;1.704~2.928V
L6 = P150, 0.15A,default 2.96;1.8~2.96V
L7 = P600, 0.15A,default off;2.856~3.104V
L8 = P150, 0.15A,default off;1.8V
L9 = P600, 0.6A,default 2.96;2.704~2.96V
L10 = P600, 0.6A,default off;3~3.312V
L11 = P600, 0.6A,default off;3~3.312V



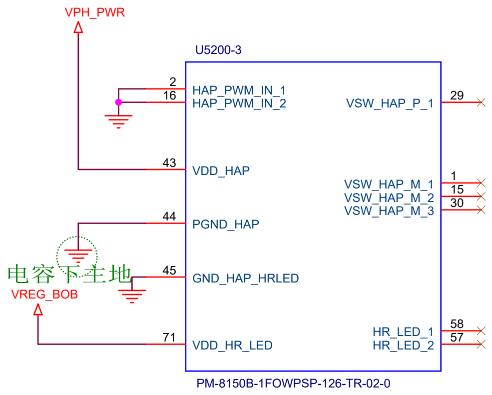
Title	K81A_TEST_BOARD	
Size A4	Document Number PM855A LDOs	Rev
Date:	Thursday, May 27, 2021	Sheet



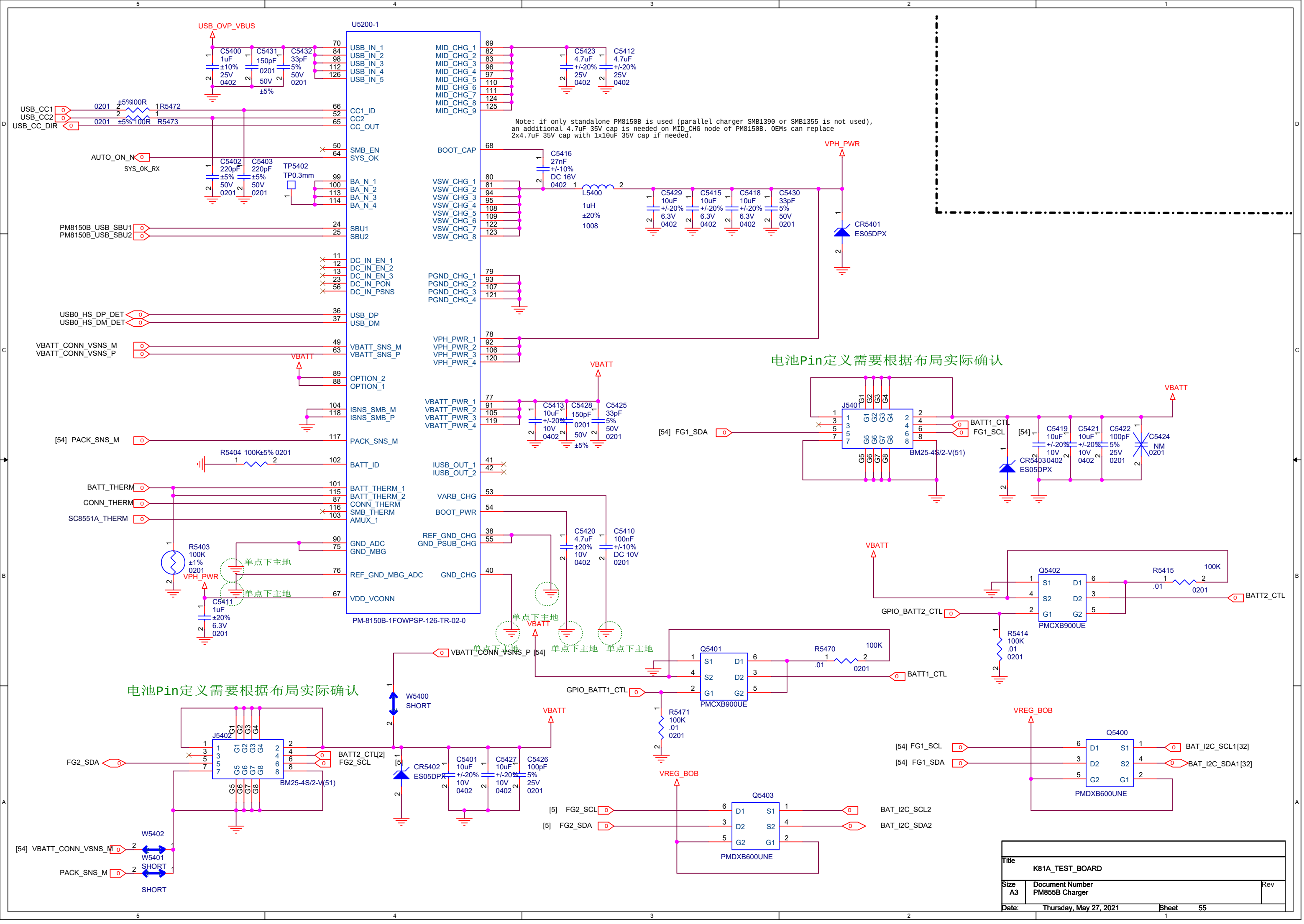
Note: GPIO_11 of PM8150B is for internal usage. Keep it float.

Note: GPIO_04 and GPIO_11 of PM8150B are for internal usage. Keep them float.
Keep GPIO_03 of PM8150B float. Don't use it for other purpose.
GPIO_06 of PM8150B is dedicated for SMB_STAT. Don't use it for other purpose.

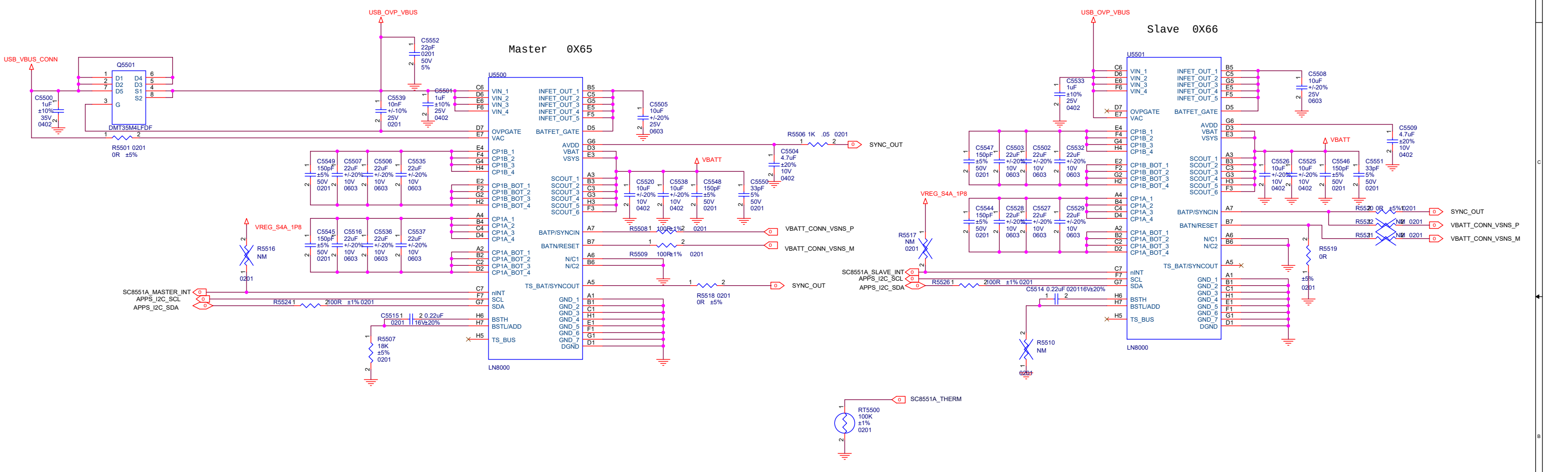
Title	K81A_TEST_BOARD		
Size A4	Document Number PM855B CTRL&GND		Rev
Date:	Thursday, May 27, 2021	Sheet	53



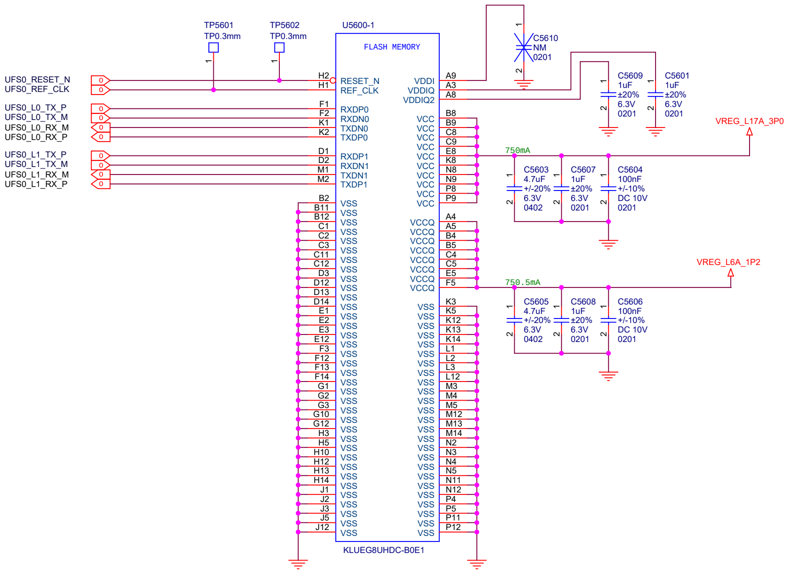
Title	K81A_TEST_BOARD	
Size A4	Document Number PM855B HAPTIC	Rev
Date:	Thursday, May 27, 2021	Sheet



USB_OVP_VBUS
U5200-1
USB_IN_1
USB_IN_2
USB_IN_3
USB_IN_4
USB_IN_5
MID_CHG_1
MID_CHG_2
MID_CHG_3
MID_CHG_4
MID_CHG_5
MID_CHG_6
MID_CHG_7
MID_CHG_8
MID_CHG_9
C5400 1uF ±10% 25V 0402
C5431 150pF 0201 50V ±5%
C5432 33pF 5% 50V 0201
C5423 4.7uF +/-20% 25V 0402
C5412 4.7uF +/-20% 25V 0402
USB_CC1
USB_CC2
USB_CC_DIR
R5472
R5473
CC1_ID
CC2
CC_OUT
Note: if only standalone PM8150B is used (parallel charger SMB1390 or SMB1355 is not used), an additional 4.7uF 35V cap is needed on MID_CHG node of PM8150B. OEMs can replace 2x4.7uF 35V cap with 1x10uF 35V cap if needed.
AUTO_ON_N
SYS_OK_RX
SMB_EN
SYS_OK
BOOT_CAP
C5416 27nF +/-10% DC 16V 0402
VPH_PWR
C5402 220pF ±5% 50V 0201
C5403 220pF ±5% 50V 0201
TP5402 TP0.3mm
BA_N_1
BA_N_2
BA_N_3
BA_N_4
VSW_CHG_1
VSW_CHG_2
VSW_CHG_3
VSW_CHG_4
VSW_CHG_5
VSW_CHG_6
VSW_CHG_7
VSW_CHG_8
L5400 1uH ±20% 1008
C5429 10uF +/-20% 6.3V 0402
C5415 10uF +/-20% 6.3V 0402
C5418 10uF +/-20% 6.3V 0402
C5430 33pF 5% 50V 0201
CR5401 ES05DPX
PM8150B_USB_SBU1
PM8150B_USB_SBU2
SBU1
SBU2
DC_IN_EN_1
DC_IN_EN_2
DC_IN_EN_3
DC_IN_PON
DC_IN_PSNS
PGND_CHG_1
PGND_CHG_2
PGND_CHG_3
PGND_CHG_4
USB0_HS_DP_DET
USB0_HS_DM_DET
USB_DP
USB_DM
VBATT_CONN_VSNS_M
VBATT_CONN_VSNS_P
VBATT_SNS_M
VBATT_SNS_P
VPH_PWR_1
VPH_PWR_2
VPH_PWR_3
VPH_PWR_4
VBATT
电池Pin定义需要根据布局实际确认
OPTION_2
OPTION_1
ISNS_SMB_M
ISNS_SMB_P
VBATT_PWR_1
VBATT_PWR_2
VBATT_PWR_3
VBATT_PWR_4
C5413 10uF +/-20% 10V 0402
C5428 150pF 0201 50V ±5%
C5425 33pF 5% 50V 0201
J5401
BATT1_CTL
FG1_SCL
[54] FG1_SDA
BM25-4S/2-V(51)
C5419 10uF +/-20% 10V 0402
C5421 10uF +/-20% 10V 0402
C5422 100pF 5% 25V 0201
C5424 NM 0201
CR5403 ES05DPX
[54] PACK_SNS_M
PACK_SNS_M
R5404 100K ±5% 0201
BATT_ID
IUSB_OUT_1
IUSB_OUT_2
BATT_THERM
CONN_THERM
SC8551A_THERM
BATT_THERM_1
BATT_THERM_2
CONN_THERM
SMB_THERM
AMUX_1
VARB_CHG
BOOT_PWR
R5403 100K ±1% 0201
REF_GND_CHG
GND_PSUB_CHG
GND_ADC
GND_MBG
C5420 4.7uF ±20% 10V 0402
C5410 100nF +/-10% DC 10V 0201
单点下主地
REF_GND_MBG_ADC
GND_CHG
C5411 1uF ±20% 6.3V 0201
VDD_VCONN
PM-8150B-1FOWPSP-126-TR-02-0
Q5402
R5415 100K .01 0201
BATT2_CTL
GPIO_BATT2_CTL
PMCXB900UE
R5414 100K .01 0201
Q5401
R5470 100K .01 0201
BATT1_CTL
GPIO_BATT1_CTL
R5471 100K .01 0201
W5400 SHORT
J5402
BATT2_CTL[2]
FG2_SCL
FG2_SDA
CR5402 ES05DPX
C5401 10uF +/-20% 10V 0402
C5427 10uF +/-20% 10V 0402
C5426 100pF 5% 25V 0201
VREG_BOB
Q5400
[54] FG1_SCL
[54] FG1_SDA
BAT_I2C_SCL1[32]
BAT_I2C_SDA1[32]
PMDXB600UNE
Q5403
[5] FG2_SCL
[5] FG2_SDA
BAT_I2C_SCL2
BAT_I2C_SDA2
W5402
[54] VBATT_CONN_VSNS_M
W5401 SHORT
SHORT
Title K81A_TEST_BOARD
Size A3
Document Number PM855B Charger
Rev
Date: Thursday, May 27, 2021
Sheet 55



Title	K81A_TEST_BOARD	
Size A2	Document Number SMB1396	Rev
Date: Friday, May 28, 2021	Sheet	

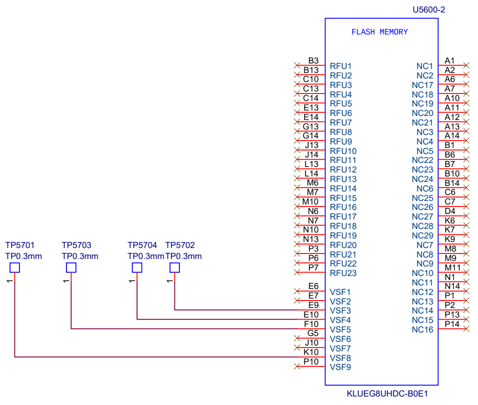


项目	配置	小米料号	P/N	供应商	描述	一供/二供	制程
K81(870)	6+128 8+256	276300000B79	KLUDG4UHDC-B0E1	Samsung	128GB UFS 3.1	First Source	V6
		276300000UJY	THGJFCT0T44BAKL	Kioxia	128GB UFS 3.0	Second Source	96L
		276300000C79	KLUEG8UHDC-B0E1	Samsung	256GB UFS 3.1	First Source	V6
		276100000H52	MT62F768M64D4BG-031 WT:A	Micron	6GB LPDDR5 for 865	First Source	1ynm
		276100000J79	K3LK7K70BM-BGCP	Samsung	8GB LPDDR5 for 875	First Source	1znm
		276100000P52	MT62F1G64D8CH-031 WT:B	Micron	8GB LPDDR5 for 875	Second Source	1znm

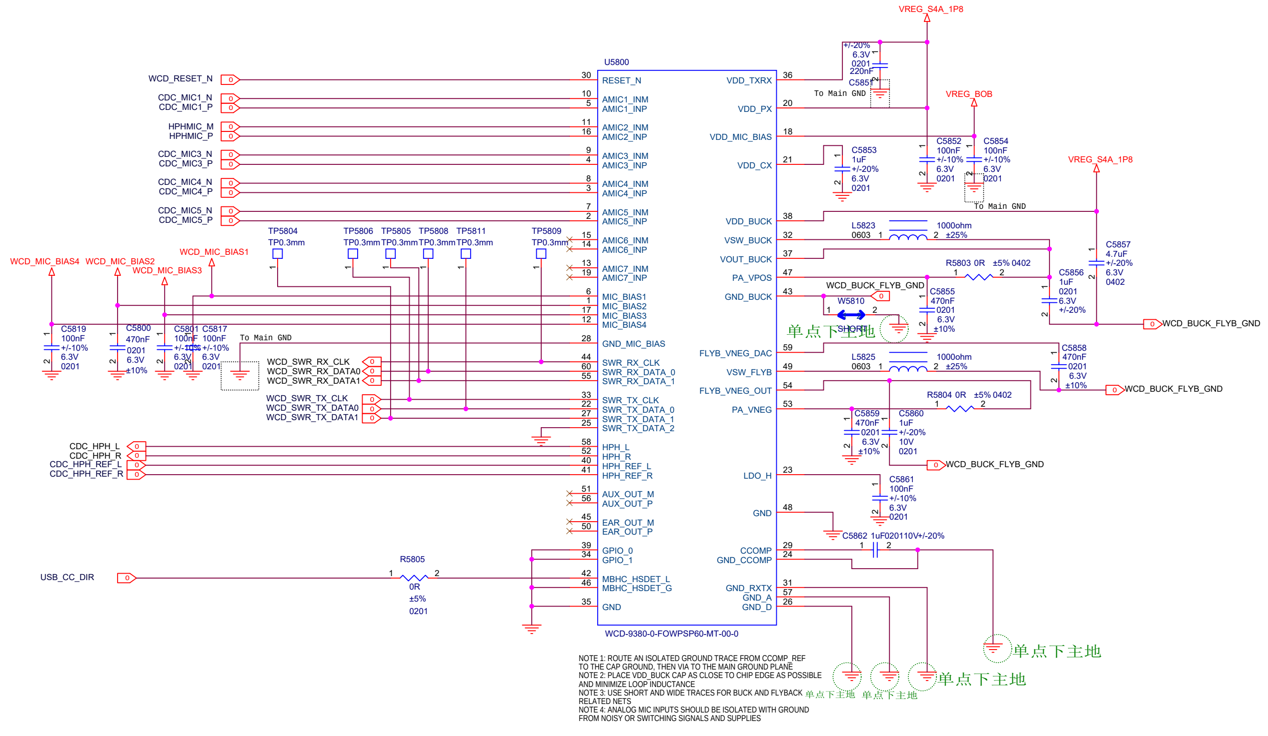
Title
K81A_TEST_BOARD

Size A3 | Document Number MEMORY | Rev

Date: Thursday, May 27, 2021 | Sheet 56 of



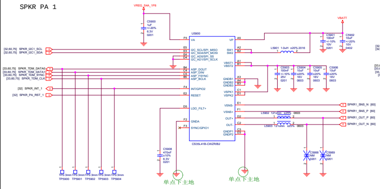
Title	K81A_TEST_BOARD		
Size A3	Document Number MEMORY-2		Rev
Date:	Thursday, May 27, 2021	Sheet 57	of



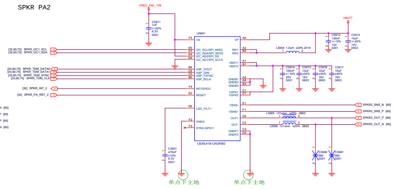
NOTE 1: ROUTE AN ISOLATED GROUND TRACE FROM CCOMP_REF TO THE CAP GROUND, THEN VIA TO THE MAIN GROUND PLANE
NOTE 2: PLACE VDD_BUCK CAP AS CLOSE TO CHIP EDGE AS POSSIBLE AND MINIMIZE LOOP INDUCTANCE
NOTE 3: USE SHORT AND WIDE TRACES FOR BUCK AND FLYBACK RELATED NETS
NOTE 4: ANALOG MIC INPUTS SHOULD BE ISOLATED WITH GROUND FROM NOISY OR SWITCHING SIGNALS AND SUPPLIES

Title	K81A_TEST_BOARD
Size A3	Document Number WCD9340-1	Rev
Date: Thursday, May 27, 2021	Sheet 58 of

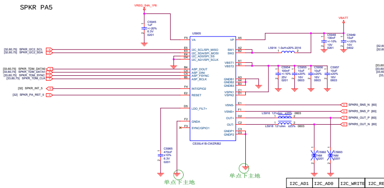
SPKR PA 1



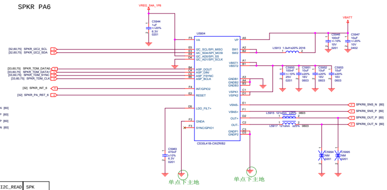
SPKR PA2



SPKR PA5



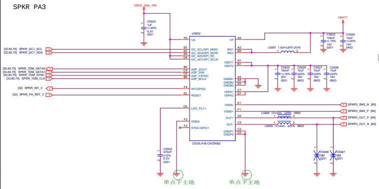
SPKR PA6



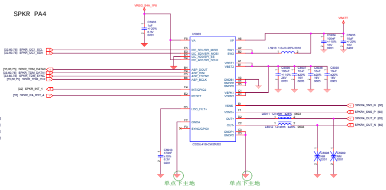
I2C_AD1	I2C_AD0	I2C_WRITE	I2C_READ	SPK
GND	GND	0x80	0x81	SPK1
GND	VA	0x82	0x83	SPK4
VA	GND	0x84	0x85	SPK2
VA	VA	0x86	0x87	SPK3

I2C_AD1	I2C_AD0	I2C_WRITE	I2C_READ	SPK
GND	GND	0x80	0x81	SPK1
GND	VA	0x82	0x83	SPK4
VA	GND	0x84	0x85	SPK2
VA	VA	0x86	0x87	SPK3

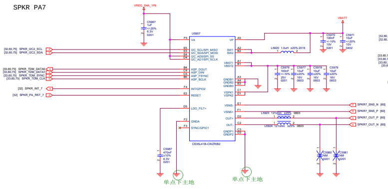
SPKR PA3



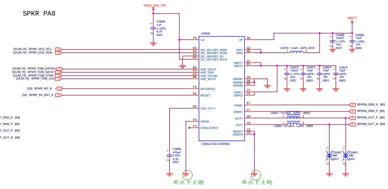
SPKR PA4

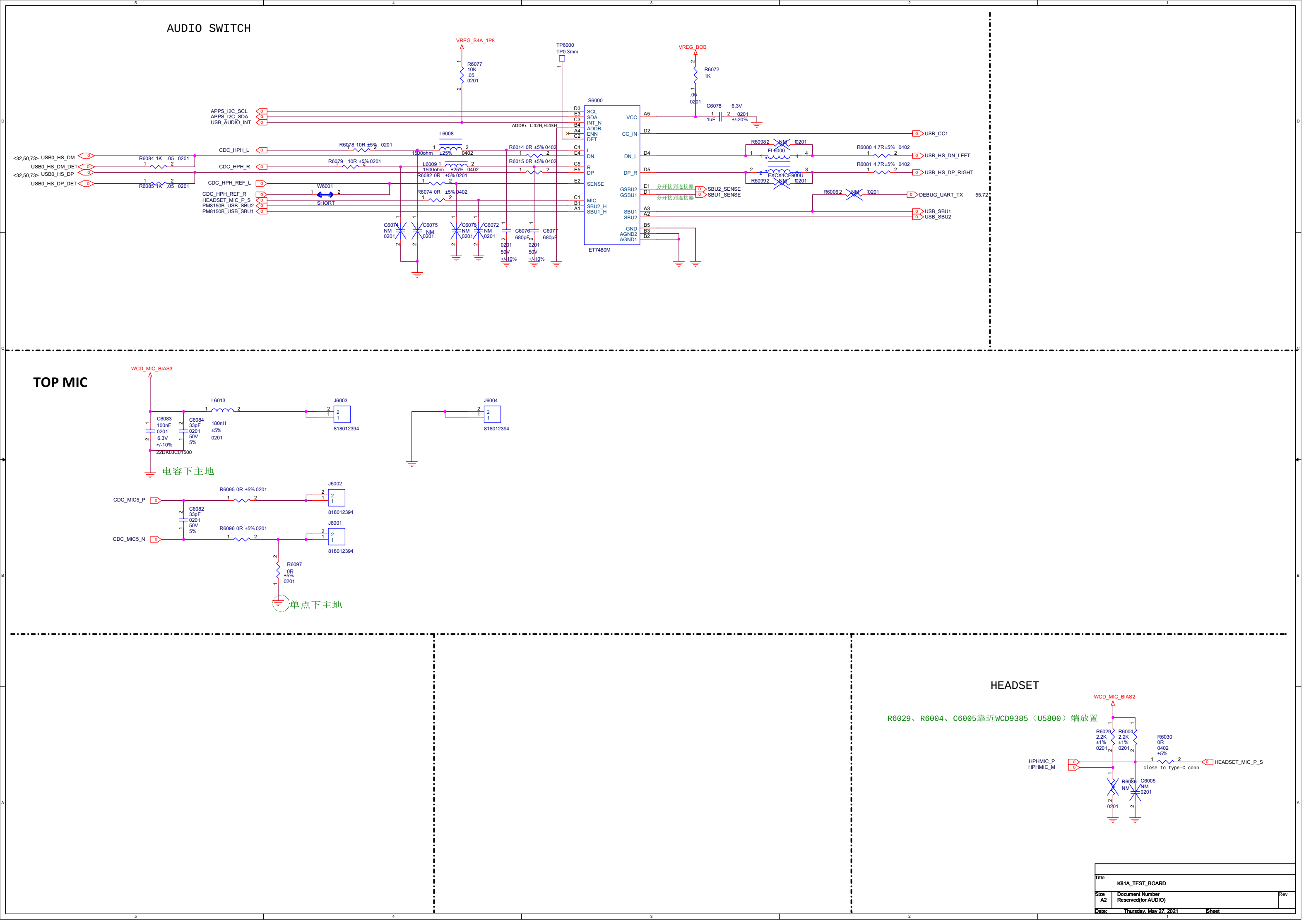


SPKR PA7

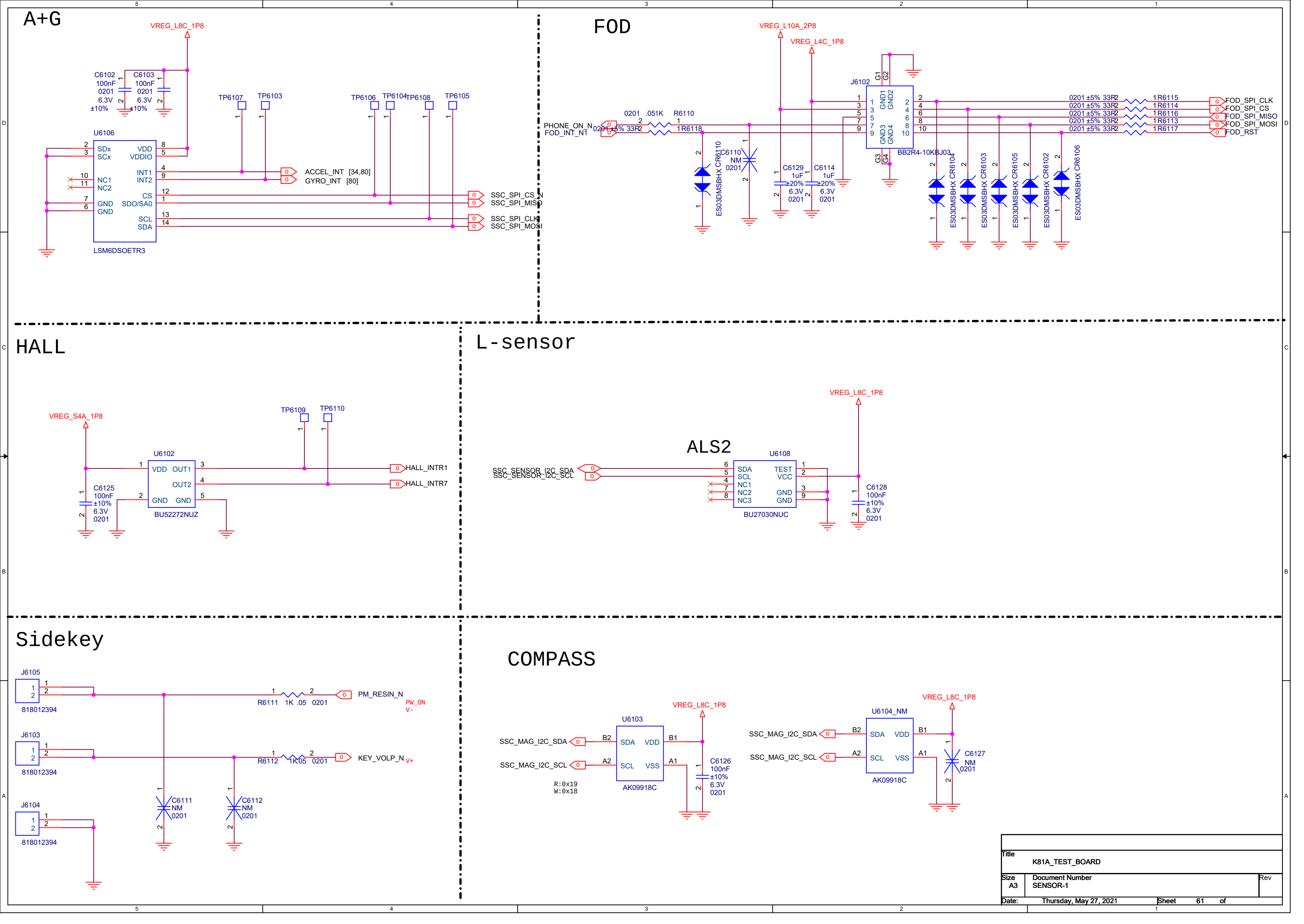


SPKR PA8

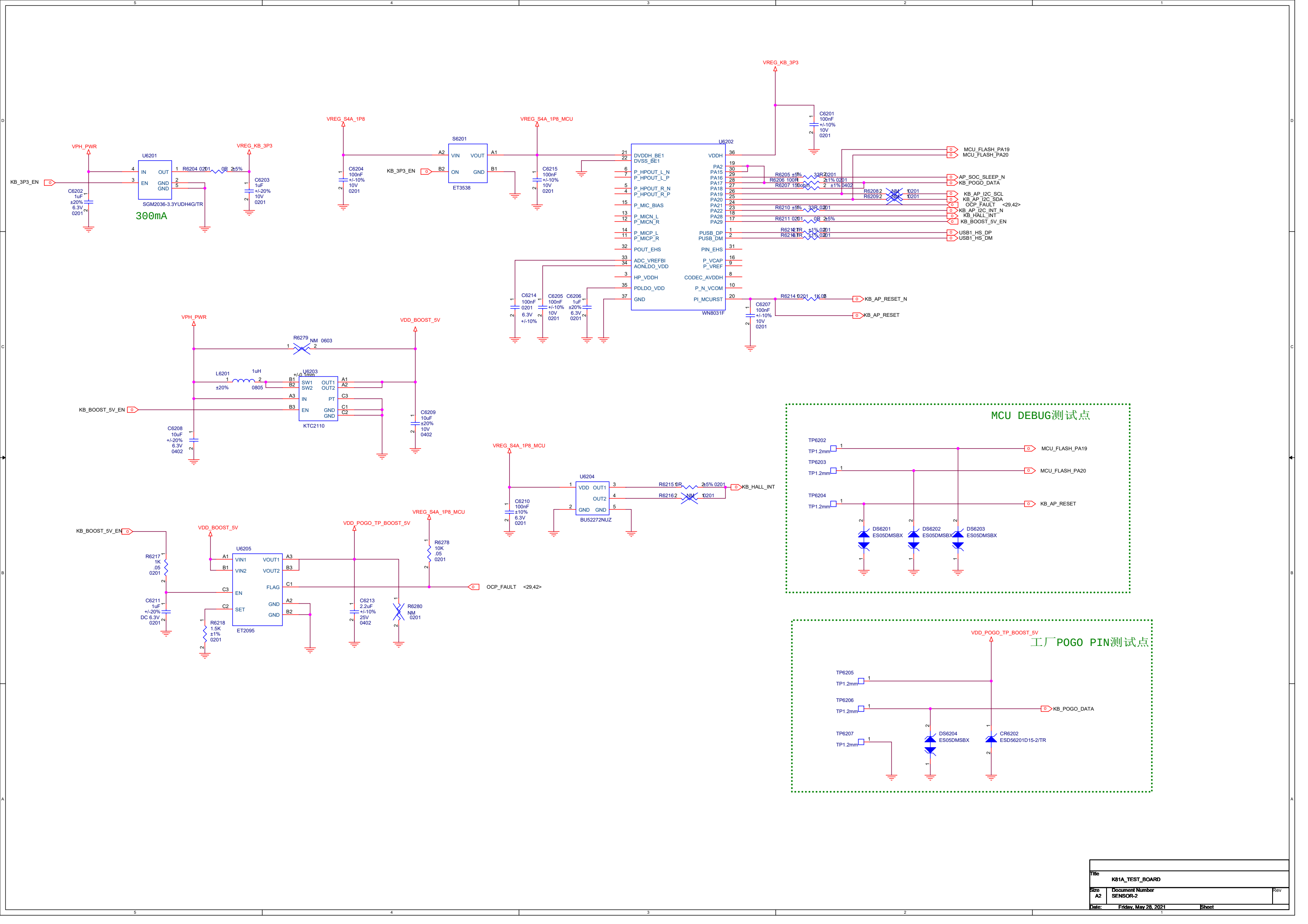


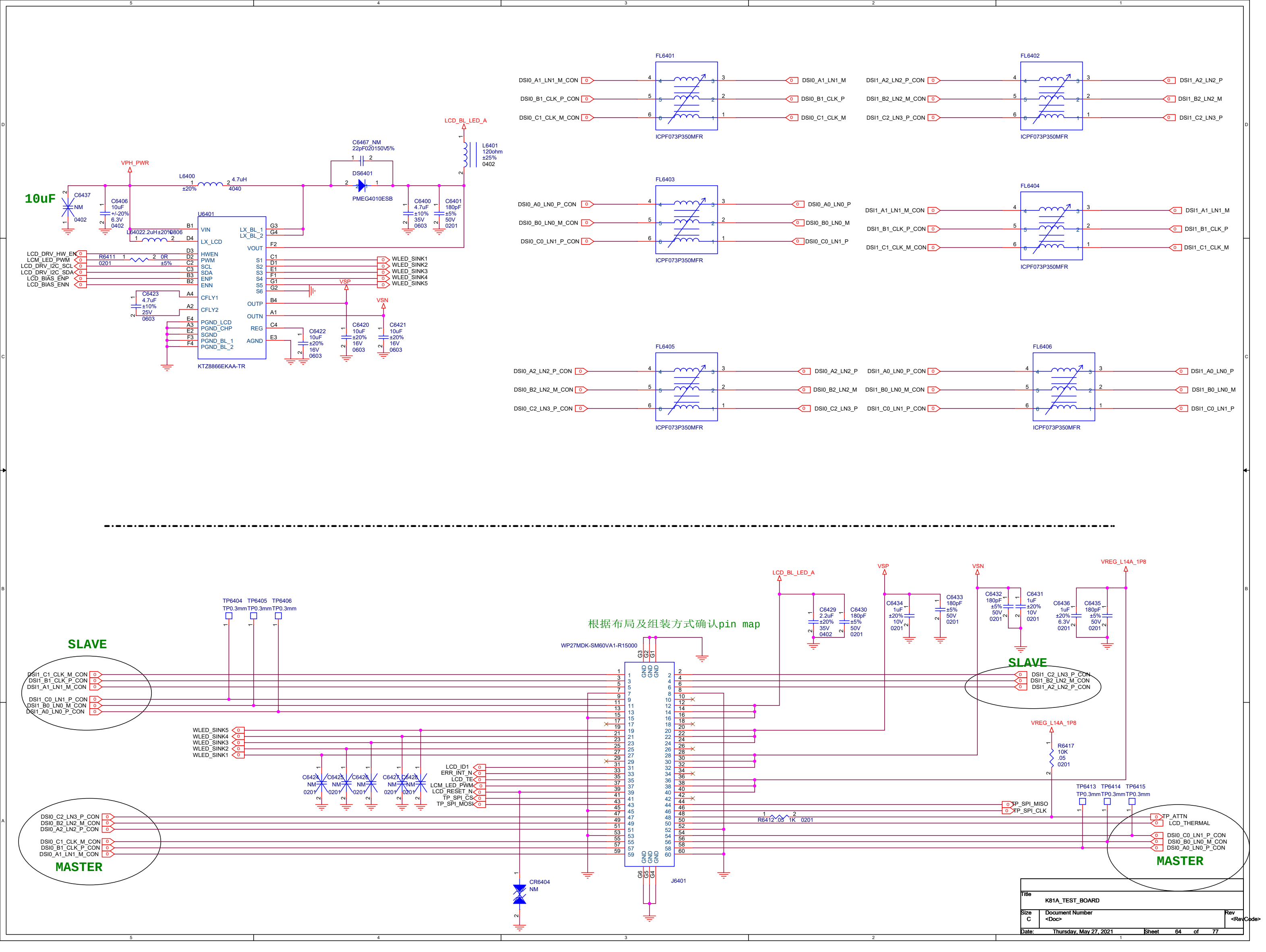


AUDIO SWITCH
VREG_S4A_1P8
R6077
10K
.05
0201
TP6000
TP0.3mm
VREG_BOB
R6072
1K
S6000
ET7480M
SCL
SDA
INT_N
ADDR
ENN
DET
L
DN
R
DP
SENSE
MIC
SBU2_H
SBU1_H
VCC
CC_IN
DN_L
DP_R
GSBU2
GSBU1
SBU1
SBU2
GND
AGND2
AGND1
ADDR：L:42H,H:43H
APPS_I2C_SCL
APPS_I2C_SDA
USB_AUDIO_INT
CDC_HPH_L
CDC_HPH_R
CDC_HPH_REF_L
CDC_HPH_REF_R
HEADSET_MIC_P_S
PM8150B_USB_SBU2
PM8150B_USB_SBU1
<32,50,73> USB0_HS_DM
USB0_HS_DM_DET
<32,50,73> USB0_HS_DP
USB0_HS_DP_DET
R6084 1K .05 0201
R6085 1K .05 0201
R6078 10R ±5% 0201
R6079 10R ±5% 0201
L6008
L6009
1500ohm ±25% 0402
R6014 0R ±5% 0402
R6015 0R ±5% 0402
R6082 0R ±5% 0201
R6074 0R ±5% 0402
W6001
SHORT
C6078 1uF 6.3V 0201 +/-20%
C6076 680pF 0201 50V +/-10%
C6077 680pF 0201 50V +/-10%
C6073 NM 0201
C6075 NM 0201
C6071 NM 0201
C6072 NM 0201
USB_CC1
R6098 NM 0201
R6099 NM 0201
FL6000
EXCX4CE900U
R6080 4.7R ±5% 0402
R6081 4.7R ±5% 0402
USB_HS_DN_LEFT
USB_HS_DP_RIGHT
R6006 NM 0201
DEBUG_UART_TX
55,72
分开接到连接器
SBU2_SENSE
SBU1_SENSE
USB_SBU1
USB_SBU2
TOP MIC
WCD_MIC_BIAS3
L6013
180nH
±5%
0201
C6083 100nF 0201 6.3V +/-10%
C6084 33pF 0201 50V 5%
22DK03C01500
电容下主地
J6003
818012394
J6004
818012394
CDC_MIC5_P
CDC_MIC5_N
R6095 0R ±5% 0201
R6096 0R ±5% 0201
C6082 33pF 0201 50V 5%
J6002
818012394
J6001
818012394
R6097 0R ±5% 0201
单点下主地
HEADSET
WCD_MIC_BIAS2
R6029、R6004、C6005靠近WCD9385（U5800）端放置
R6029 2.2K ±1% 0201
R6004 2.2K ±1% 0201
R6030 0R 0402 ±5%
HPHMIC_P
HPHMIC_M
HEADSET_MIC_P_S
close to type-C conn
R6036 NM 0201
C6005 NM 0201
Title
K81A_TEST_BOARD
Size
A2
Document Number
Reserved(for AUDIO)
Rev
Date:
Thursday, May 27, 2021
Sheet

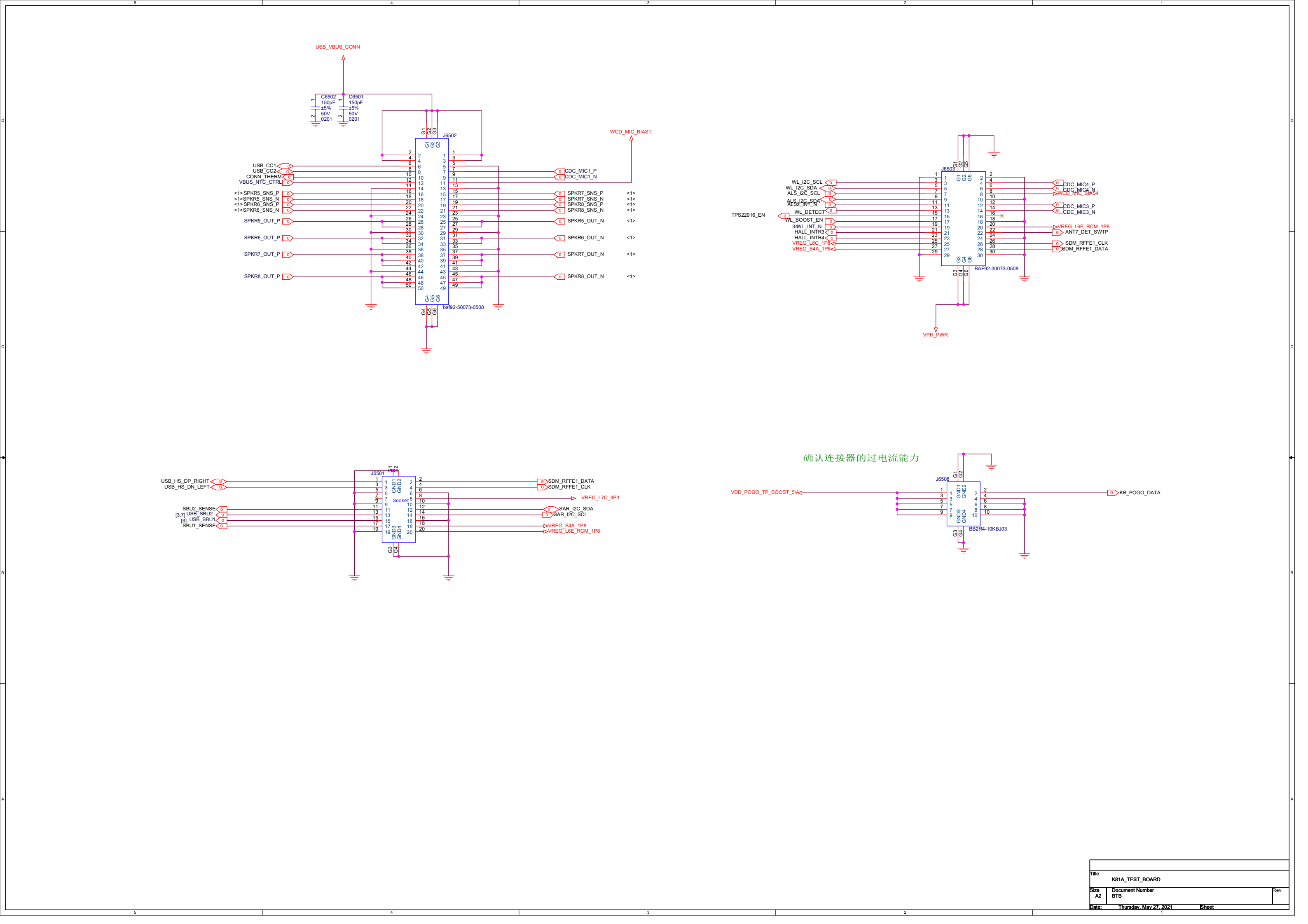


Title	K81A_TEST_BOARD
Size A3	Document Number SENSOR-1
Rev	
Date: Thursday, May 27, 2021	Sheet 61 of

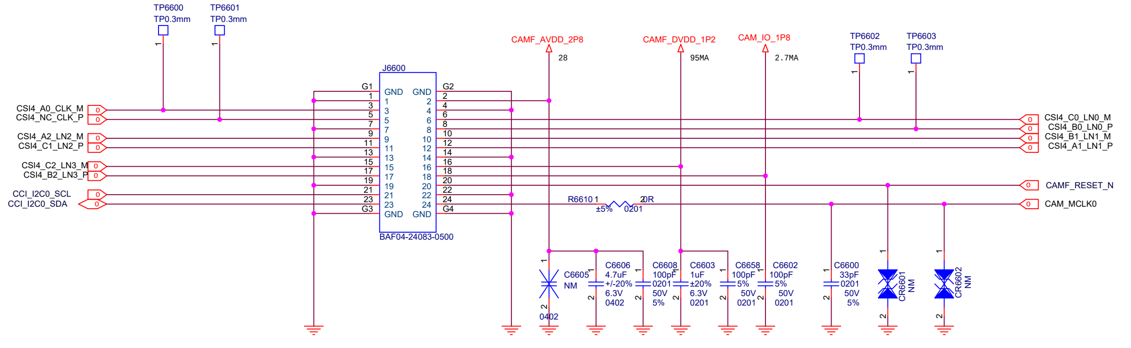

VREG_KB_3P3
VPH_PWR
KB_3P3_EN
U6201
SGM2036-3.3YUDH4G/TR
300mA
VREG_S4A_1P8
S6201
ET3538
KB_3P3_EN
VREG_S4A_1P8_MCU
U6202
WN8031F
MCU_FLASH_PA19
MCU_FLASH_PA20
AP_SOC_SLEEP_N
KB_POGO_DATA
KB_AP_I2C_SCL
KB_AP_I2C_SDA
OCP_FAULT <29,42>
KB_AP_I2C_INT_N
KB_HALL_INT
KB_BOOST_5V_EN
USB1_HS_DP
USB1_HS_DM
KB_AP_RESET_N
KB_AP_RESET
VDD_BOOST_5V
U6203
KTC2110
KB_BOOST_5V_EN
U6204
BU52272NUZ
KB_HALL_INT
MCU DEBUG测试点
TP6202
TP6203
TP6204
MCU_FLASH_PA19
MCU_FLASH_PA20
KB_AP_RESET
DS6201 ES05DMSBX
DS6202 ES05DMSBX
DS6203 ES05DMSBX
VDD_POGO_TP_BOOST_5V
U6205
ET2095
OCP_FAULT <29,42>
工厂POGO PIN测试点
TP6205
TP6206
TP6207
KB_POGO_DATA
DS6204 ES05DMSBX
CR6202 ESD56201D15-2/TR
Title K81A_TEST_BOARD
Size A2
Document Number SENSOR-2
Rev
Date: Friday, May 28, 2021
Sheet



10uF
C6437 NM 0402
C6406 10uF +/-20% 6.3V 0402
VPH_PWR
L6400 ±20% 4.7uH 4040
C6467_NM 22pF020150V5%
DS6401 PMEG4010ESB
LCD_BL_LED_A
L6401 120ohm ±25% 0402
C6400 4.7uF ±10% 35V 0603
C6401 180pF ±5% 50V 0201
U6401
KTZ8866EKAA-TR
VIN
LX_LCD
HWEN
PWM
SCL
SDA
ENP
ENN
CFLY1
CFLY2
PGND_LCD
PGND_CHP
SGND
PGND_BL_1
PGND_BL_2
LX_BL_1
LX_BL_2
VOUT
OUTP
OUTN
REG
AGND
LCD_DRV_HW_EN
LCM_LED_PWM
LCD_DRV_I2C_SCL
LCD_DRV_I2C_SDA
LCD_BIAS_ENP
LCD_BIAS_ENN
R6411 0R ±5% 0201
C6423 4.7uF ±10% 25V 0603
WLED_SINK1
WLED_SINK2
WLED_SINK3
WLED_SINK4
WLED_SINK5
VSP
VSN
C6420 10uF ±20% 16V 0603
C6421 10uF ±20% 16V 0603
C6422 10uF ±20% 16V 0603
FL6401
FL6402
FL6403
FL6404
FL6405
FL6406
ICPF073P350MFR
DSI0_A1_LN1_M_CON
DSI0_B1_CLK_P_CON
DSI0_C1_CLK_M_CON
DSI0_A1_LN1_M
DSI0_B1_CLK_P
DSI0_C1_CLK_M
DSI1_A2_LN2_P_CON
DSI1_B2_LN2_M_CON
DSI1_C2_LN3_P_CON
DSI1_A2_LN2_P
DSI1_B2_LN2_M
DSI1_C2_LN3_P
DSI0_A0_LN0_P_CON
DSI0_B0_LN0_M_CON
DSI0_C0_LN1_P_CON
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_C0_LN1_P
DSI1_A1_LN1_M_CON
DSI1_B1_CLK_P_CON
DSI1_C1_CLK_M_CON
DSI1_A1_LN1_M
DSI1_B1_CLK_P
DSI1_C1_CLK_M
DSI0_A2_LN2_P_CON
DSI0_B2_LN2_M_CON
DSI0_C2_LN3_P_CON
DSI0_A2_LN2_P
DSI0_B2_LN2_M
DSI0_C2_LN3_P
DSI1_A0_LN0_P_CON
DSI1_B0_LN0_M_CON
DSI1_C0_LN1_P_CON
DSI1_A0_LN0_P
DSI1_B0_LN0_M
DSI1_C0_LN1_P
根据布局及组装方式确认pin map
WP27MDK-SM60VA1-R15000
J6401
SLAVE
MASTER
TP6404 TP0.3mm
TP6405 TP0.3mm
TP6406 TP0.3mm
TP6413 TP0.3mm
TP6414 TP0.3mm
TP6415 TP0.3mm
C6429 2.2uF ±20% 35V 0402
C6430 180pF ±5% 50V 0201
C6434 1uF ±20% 10V 0201
C6433 180pF ±5% 50V 0201
C6432 180pF ±5% 50V 0201
C6431 1uF ±20% 10V 0201
C6436 1uF ±20% 6.3V 0201
C6435 180pF ±5% 50V 0201
VREG_L14A_1P8
R6417 10K .05 0201
R6412 .05 1K 0201
LCD_ID1
ERR_INT_N
LCD_TE
LCM_LED_PWM
LCD_RESET_N
TP_SPI_CS
TP_SPI_MOSI
TP_SPI_MISO
TP_SPI_CLK
TP_ATTN
LCD_THERMAL
CR6404 NM
Title K81A_TEST_BOARD
Size C
Document Number <Doc>
Rev <RevCode>
Date: Thursday, May 27, 2021
Sheet 64 of 77


USB_VBUS_CONN
C6502
150pF
±5%
50V
0201
C6501
150pF
±5%
50V
0201
J6502
baf92-50073-0508
USB_CC1
USB_CC2
CONN_THERM
VBUS_NTC_CTRL
SPKR5_SNS_P
SPKR5_SNS_N
SPKR6_SNS_P
SPKR6_SNS_N
SPKR5_OUT_P
SPKR6_OUT_P
SPKR7_OUT_P
SPKR8_OUT_P
WCD_MIC_BIAS1
CDC_MIC1_P
CDC_MIC1_N
SPKR7_SNS_P
SPKR7_SNS_N
SPKR8_SNS_P
SPKR8_SNS_N
SPKR5_OUT_N
SPKR6_OUT_N
SPKR7_OUT_N
SPKR8_OUT_N
J6503
BAF92-30073-0508
WL_I2C_SCL
WL_I2C_SDA
ALS_I2C_SCL
ALS_I2C_SDA
ALS2_INT_N
WL_DETECT
TPS22916_EN
WL_BOOST_EN
WL_INT_N
HALL_INTR3
HALL_INTR4
VREG_L8C_1P8
VREG_S4A_1P8
CDC_MIC4_P
CDC_MIC4_N
WCD_MIC_BIAS4
CDC_MIC3_P
CDC_MIC3_N
VREG_L6E_RCM_1P8
ANT7_DET_SWTP
SDM_RFFE1_CLK
SDM_RFFE1_DATA
VPH_PWR
J6501
Socket
USB_HS_DP_RIGHT
USB_HS_DN_LEFT
SBU2_SENSE
USB_SBU2
USB_SBU1
SBU1_SENSE
SDM_RFFE1_DATA
SDM_RFFE1_CLK
VREG_L7C_3P3
SAR_I2C_SDA
SAR_I2C_SCL
VREG_S4A_1P8
VREG_L6E_RCM_1P8
确认连接器的过电流能力
J6508
BB2R4-10KBJ03
VDD_POGO_TP_BOOST_5V
KB_POGO_DATA
Title
K81A_TEST_BOARD
Size
A2
Document Number
BTB
Rev
Date:
Thursday, May 27, 2021
Sheet

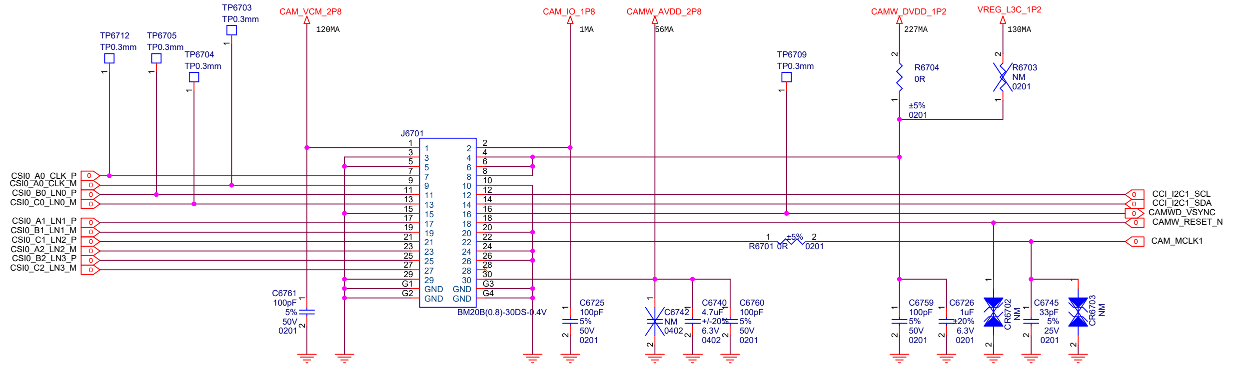
Front 8M （OV16A）



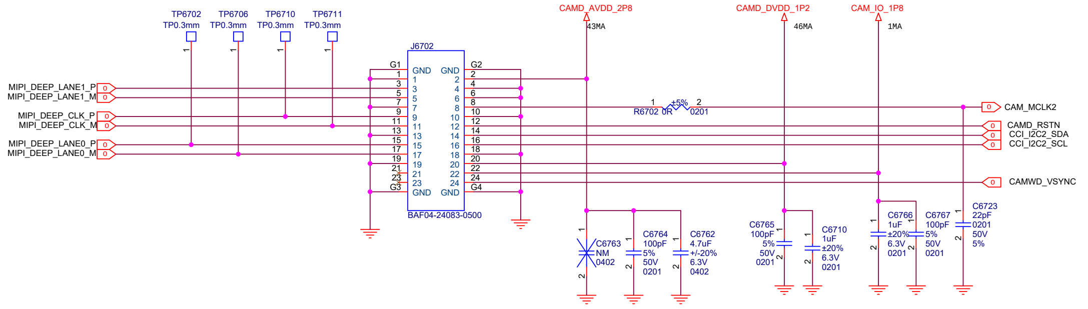
Title	K81A_TEST_BOARD	
Size A3	Document Number FRONT CAMERA	Rev
Date: Thursday, May 27, 2021	Sheet 66 of	

WIDE 50M(OV50C) FOR K81

WIDE 13M(OV13B10) FOR K81A



5M DEEP (Hi-556)



Title	K81A_TEST_BOARD	
Size A3	Document Number WIDE&TELE CAMERA	Rev
Date: Thursday, May 27, 2021	Sheet 67 of 77	

5 4 3 2 1

D

C

B

A

Title	K81A_TEST_BOARD		
Size A3	Document Number ULTRA CAMERA		Rev
Date:	Thursday, May 27, 2021	Sheet 68 of	

5 4 3 2 1

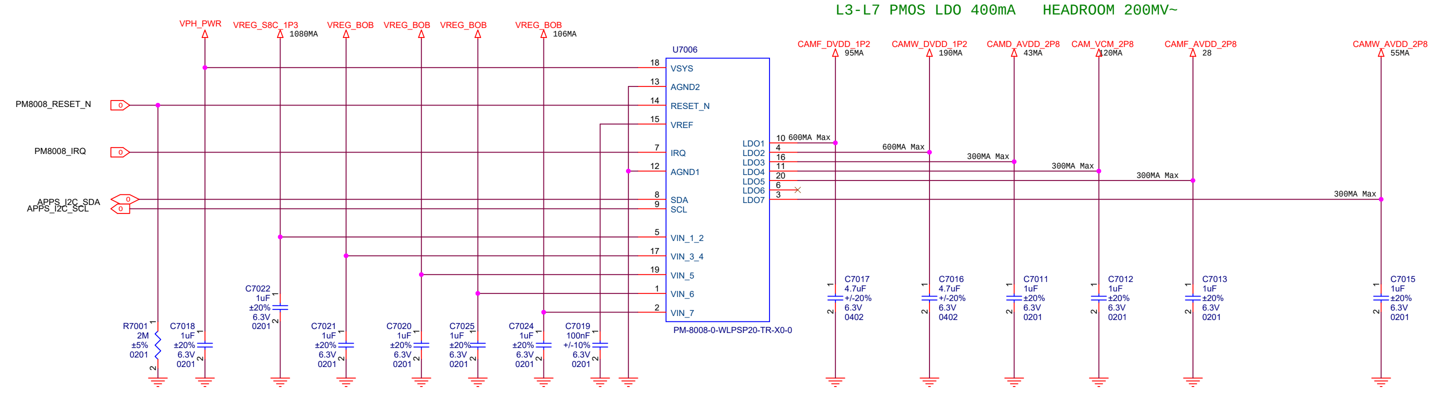
Title	K81A_TEST_BOARD		
Size A3	Document Number CAMERA LDOs		Rev
Date:	Thursday, May 27, 2021	Sheet 69	of

L1 NMOS LDO 1000mA NMOS LDO HEADROOM 200mA 75mV

L2 NMOS LDO 1000mA NMOS LDO HEADROOM 500mA 187.5mV

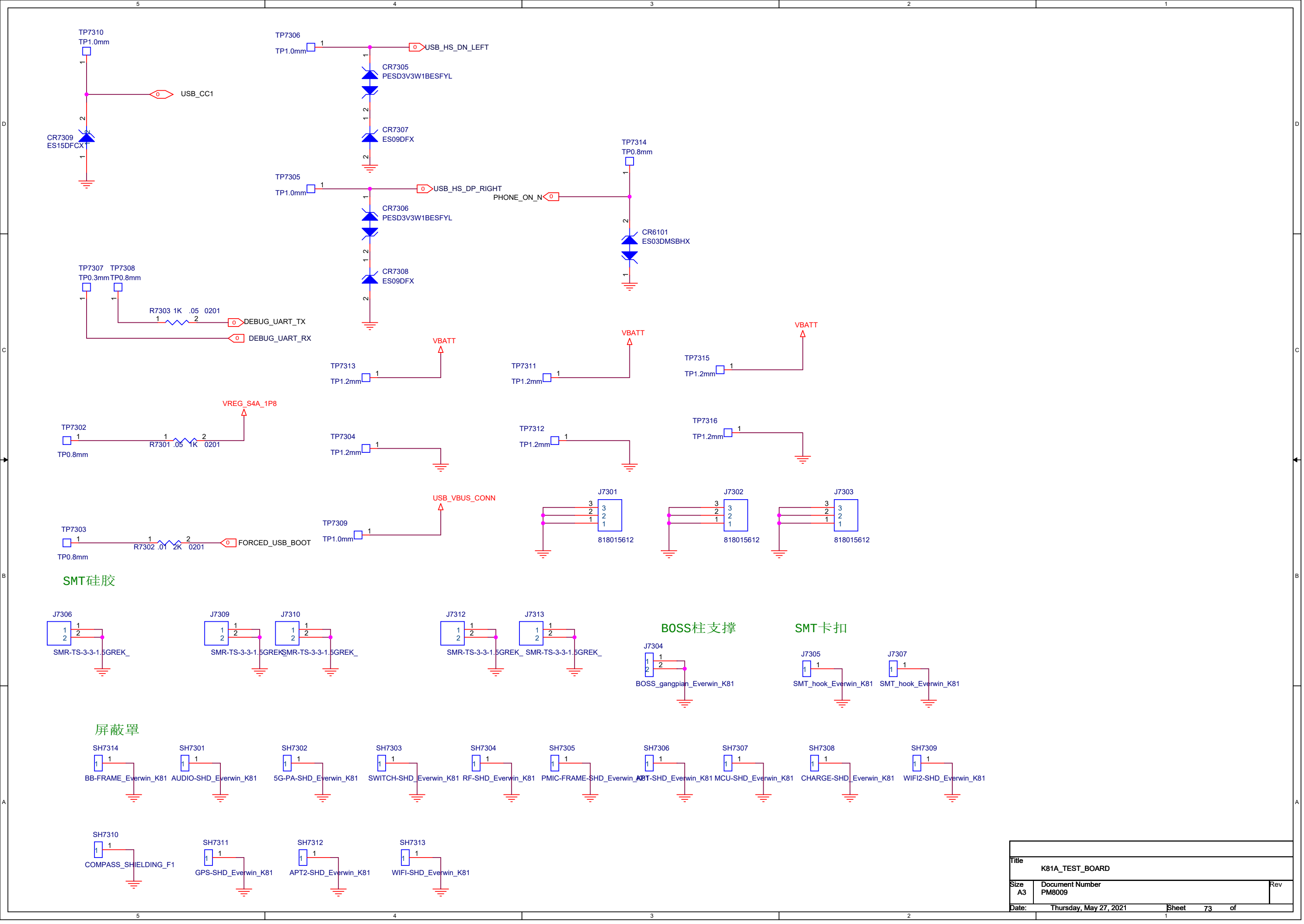
NMOS LDO HEADROOM 1000mA 200mV

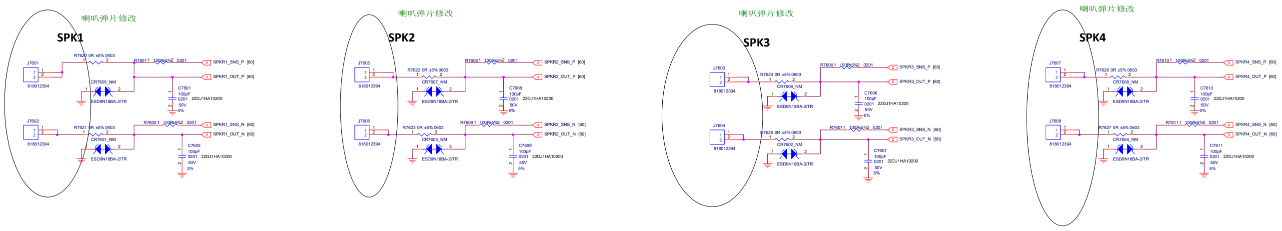
L3-L7 PMOS LDO 400mA   HEADROOM 200MV~



Title	K81A_TEST_BOARD	
Size A3	Document Number TEST POINT	Rev
Date:	Thursday, May 27, 2021	Sheet 70 of

Title	K81A_TEST_BOARD	
Size A2	Document Number USB CONNECTOR	Rev
Date:	Thursday, May 27, 2021	Sheet


TP7310
TP1.0mm
USB_CC1
CR7309
ES15DFCX
TP7306
TP1.0mm
USB_HS_DN_LEFT
CR7305
PESD3V3W1BESFYL
CR7307
ES09DFX
TP7305
TP1.0mm
USB_HS_DP_RIGHT
CR7306
PESD3V3W1BESFYL
CR7308
ES09DFX
TP7314
TP0.8mm
PHONE_ON_N
CR6101
ES03DMSBHX
TP7307
TP0.3mm
TP7308
TP0.8mm
R7303 1K .05 0201
DEBUG_UART_TX
DEBUG_UART_RX
VBATT
TP7313
TP1.2mm
TP7311
TP1.2mm
TP7315
TP1.2mm
VREG_S4A_1P8
TP7302
TP0.8mm
R7301 .05 1K 0201
TP7304
TP1.2mm
TP7312
TP1.2mm
TP7316
TP1.2mm
USB_VBUS_CONN
TP7309
TP1.0mm
J7301
J7302
J7303
818015612
TP7303
TP0.8mm
R7302 .01 2K 0201
FORCED_USB_BOOT
SMT硅胶
J7306
J7309
J7310
J7312
J7313
SMR-TS-3-3-1.5GREK_
BOSS柱支撑
J7304
BOSS_gangpian_Everwin_K81
SMT卡扣
J7305
J7307
SMT_hook_Everwin_K81
屏蔽罩
SH7314
BB-FRAME_Everwin_K81
SH7301
AUDIO-SHD_Everwin_K81
SH7302
5G-PA-SHD_Everwin_K81
SH7303
SWITCH-SHD_Everwin_K81
SH7304
RF-SHD_Everwin_K81
SH7305
PMIC-FRAME-SHD_Everwin_K81
SH7306
APT-SHD_Everwin_K81
SH7307
MCU-SHD_Everwin_K81
SH7308
CHARGE-SHD_Everwin_K81
SH7309
WIFI2-SHD_Everwin_K81
SH7310
COMPASS_SHIELDING_F1
SH7311
GPS-SHD_Everwin_K81
SH7312
APT2-SHD_Everwin_K81
SH7313
WIFI-SHD_Everwin_K81
Title
K81A_TEST_BOARD
Size
A3
Document Number
PM8009
Rev
Date:
Thursday, May 27, 2021
Sheet
73
of



Title	K81A_TEST_BOARD			
Size: D	Document Number: <Doc>			Rev: <RevCode>
Date:	Thursday, May 27, 2021	Sheet 76	of 77	